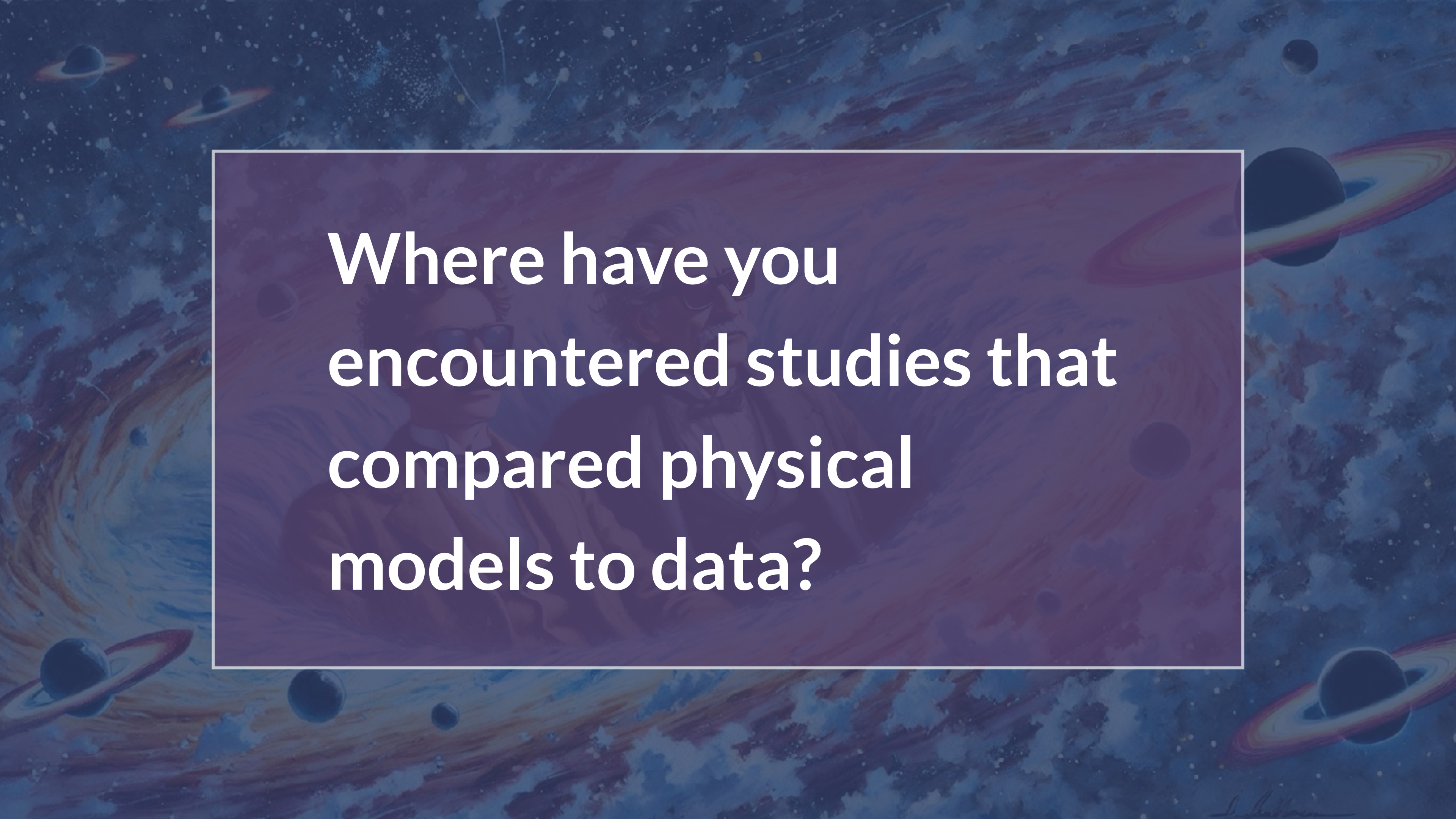


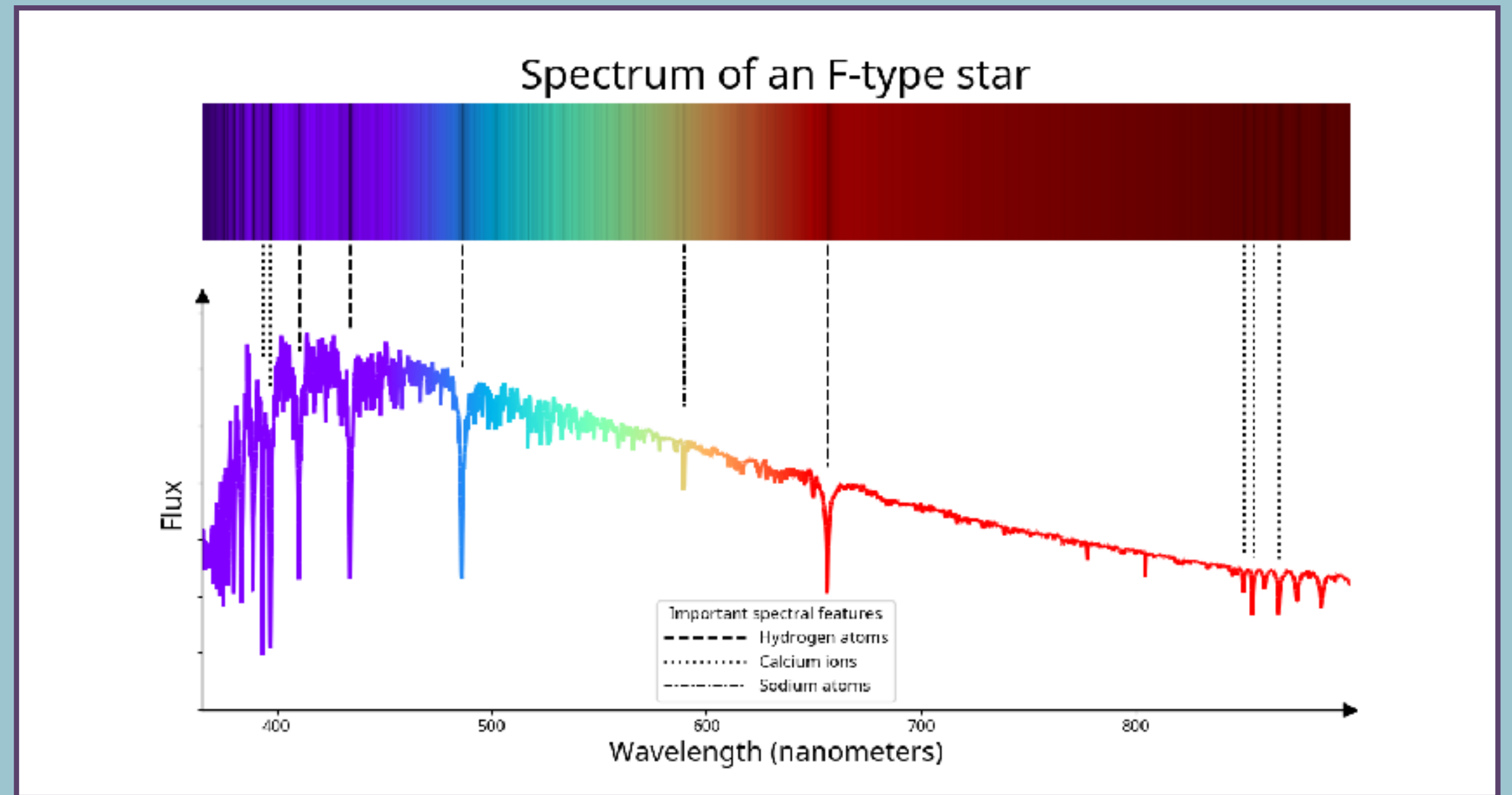


Statistical Inference I

The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies. Several planets with prominent ring systems, similar to Saturn, are visible, along with smaller moons and distant stars. The overall aesthetic is that of a classic science fiction or space exploration theme.

**Where have you
encountered studies that
compared physical
models to data?**

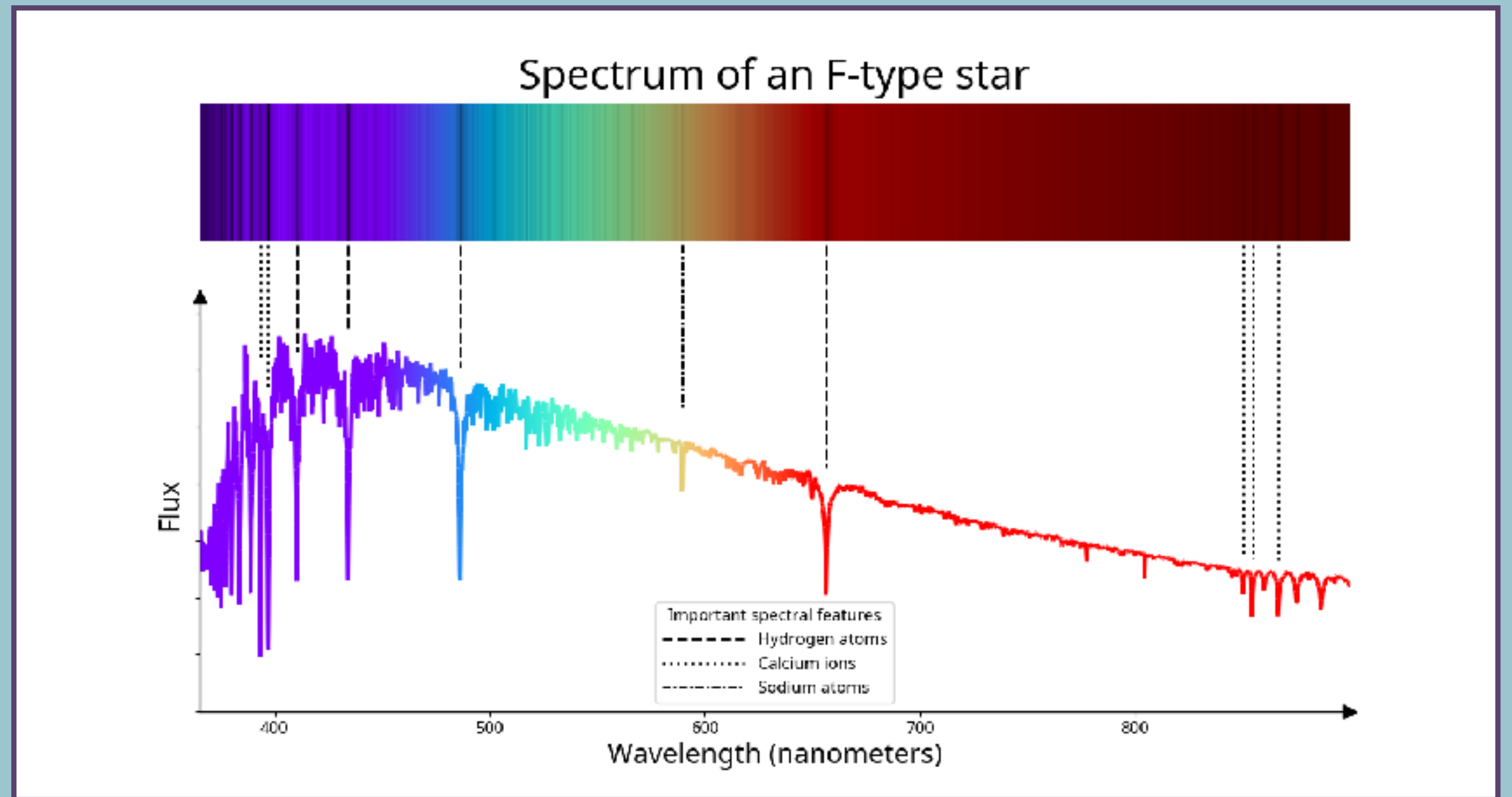
Example: estimating the temperature of a star



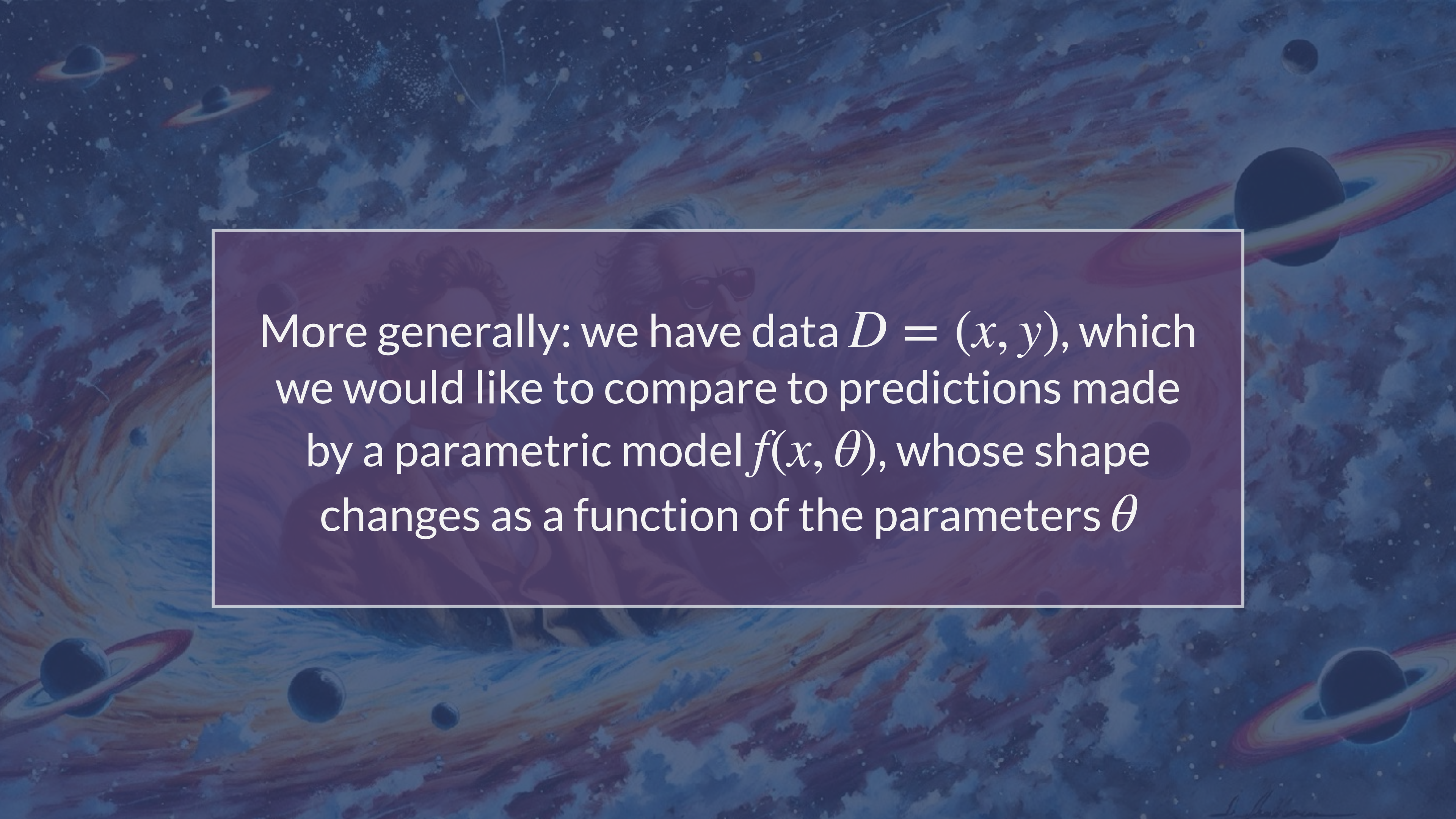
$$B_{\lambda} = \frac{2hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

Example: estimating the temperature of a star

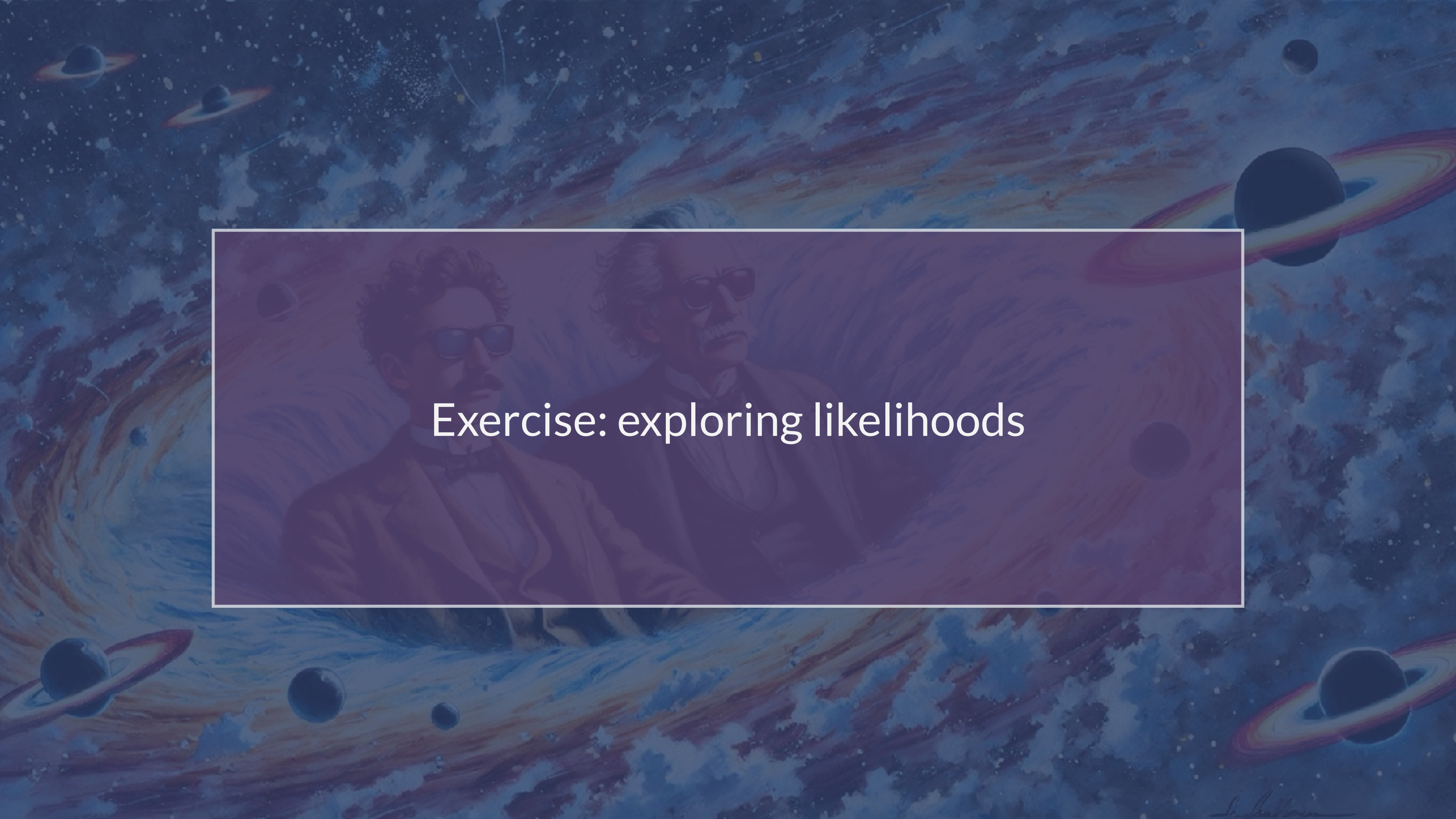
Goal: estimate the temperature T given the spectrum



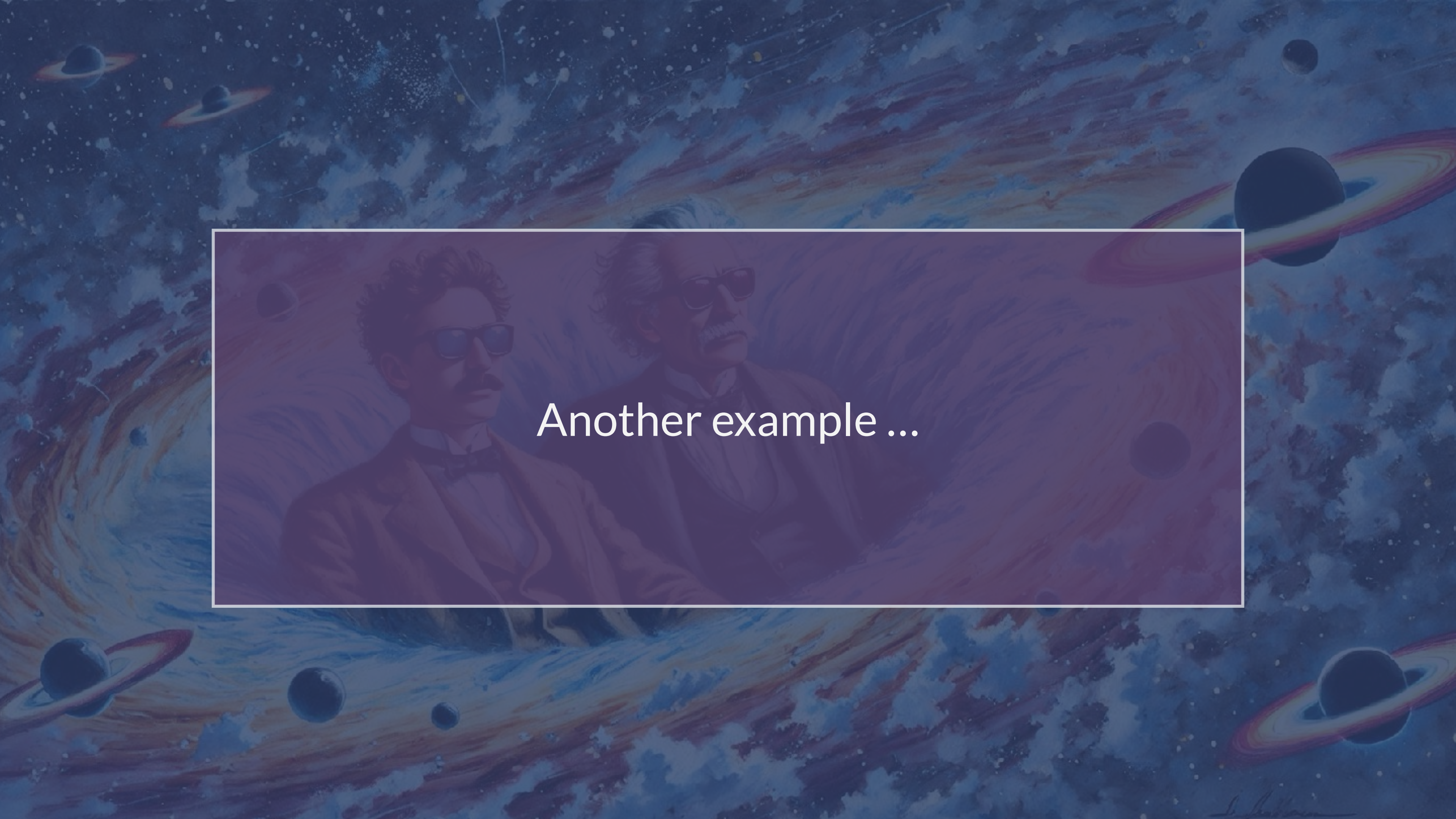
$$B_{\lambda} = \frac{2hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies including planets with rings (resembling Saturn) and smaller moons or planets. The overall tone is deep blue and purple, with a sense of vastness and cosmic wonder.

More generally: we have data $D = (x, y)$, which we would like to compare to predictions made by a parametric model $f(x, \theta)$, whose shape changes as a function of the parameters θ

The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies including planets with rings (resembling Saturn) and smaller moons or planets. The overall tone is deep blue and purple, with bright highlights from the galaxies.

Exercise: exploring likelihoods

The background of the slide is a vibrant, artistic depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies including planets with rings (resembling Saturn) and smaller moons. The overall tone is cosmic and awe-inspiring.

Another example ...

The energy dependence of the pion-proton scattering cross section

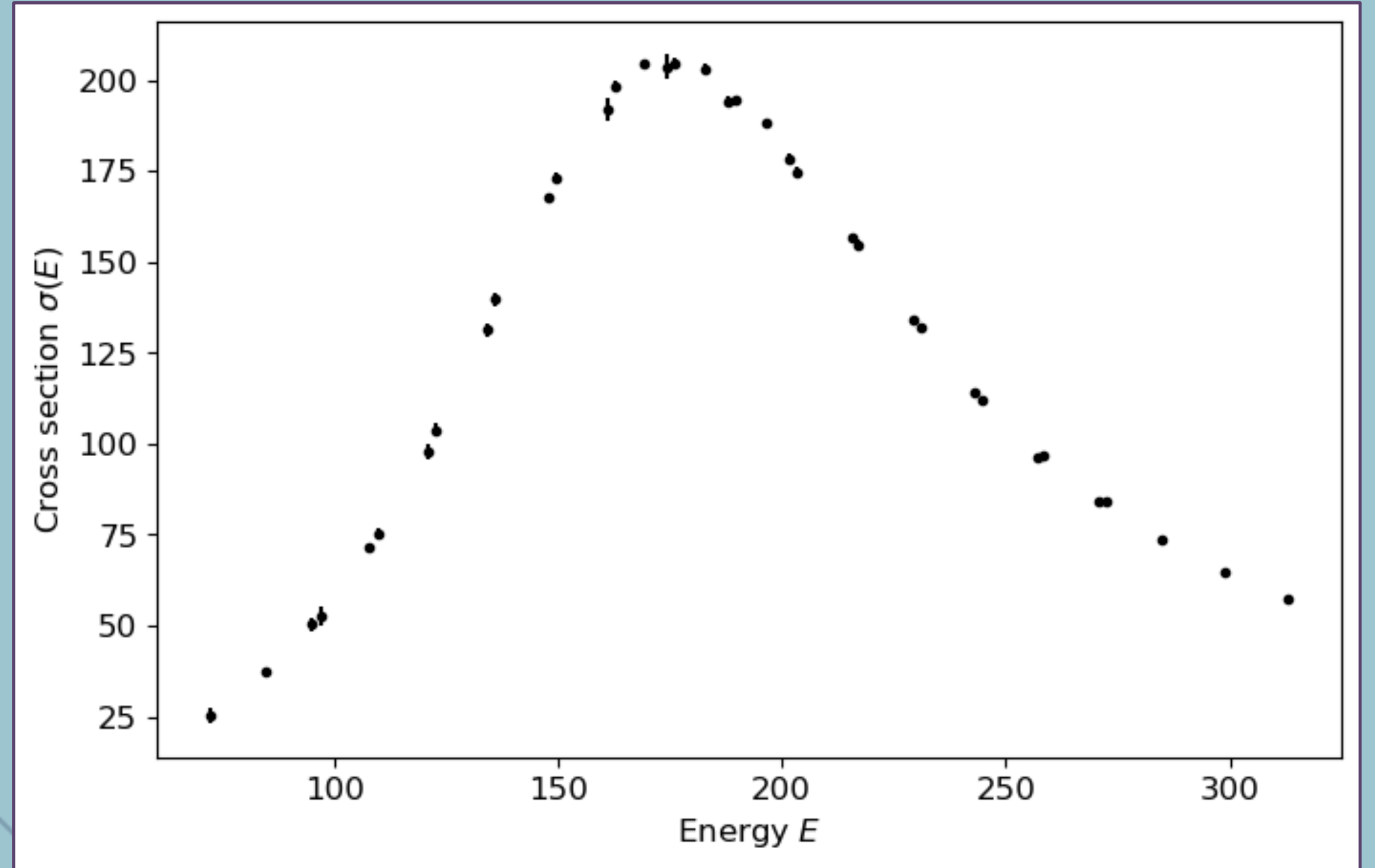
- Resonances in pion-proton interaction provide important evidence for the existence of quarks
- $\pi^+ + p \rightarrow \Delta^{++} \rightarrow \pi^+ + p$
- Δ^{++} is a short-lived resonance particle
- Goal: measure cross section of particle as a function of beam energy in MeV
- Infer resonant energy of the interaction, E_0 , by modelling the cross section with the non-relativistic Breit-Wigner formula

$$\sigma(E) = N \frac{\Gamma^2/4}{(E - E_0)^2 + \Gamma^2/4}$$

with $\Gamma(E) = \Gamma_0 \left(\frac{E}{130\text{MeV}} \right)^{1/2}$ the resonant interaction lifetime

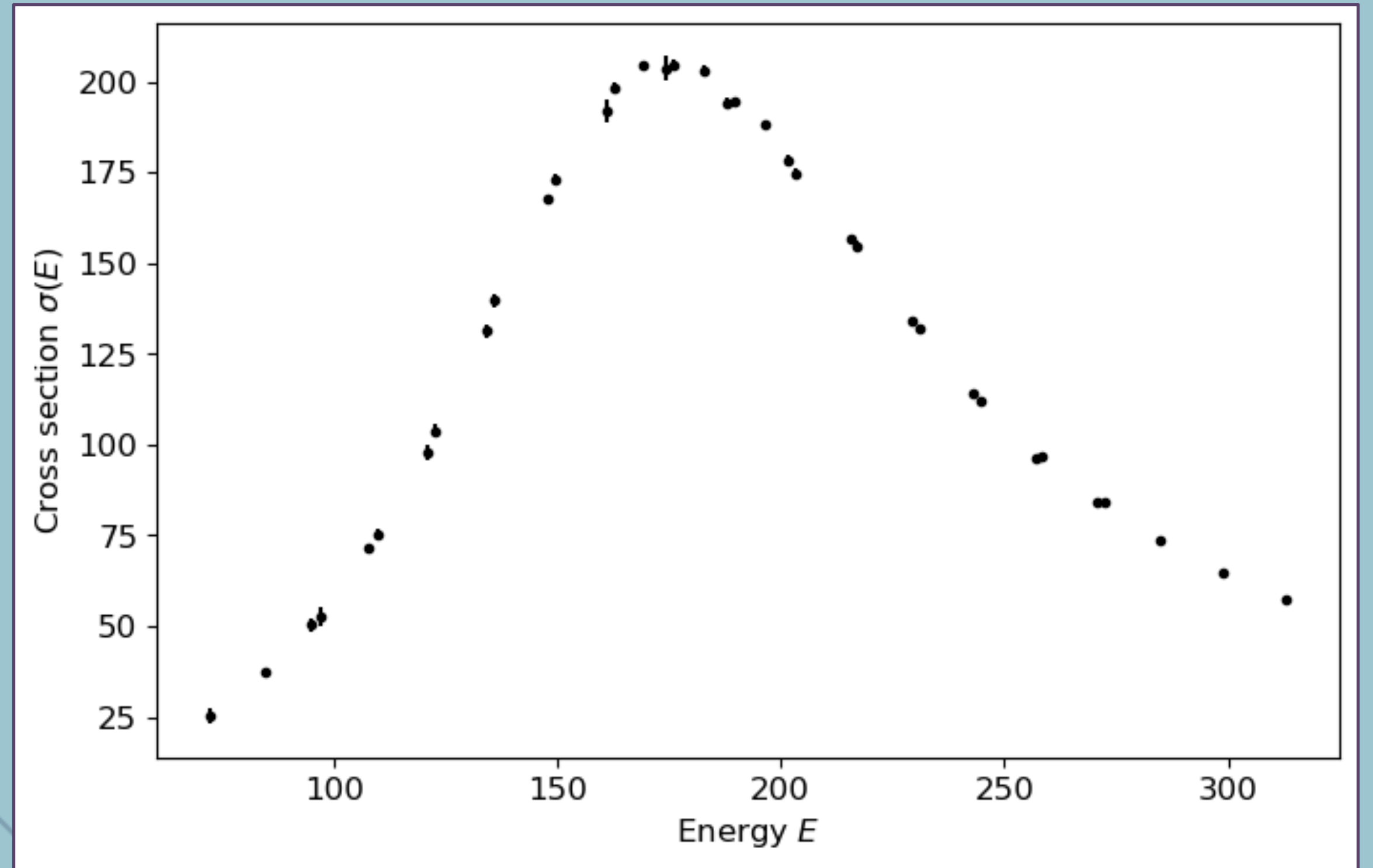
The energy dependence of the pion-proton scattering cross section

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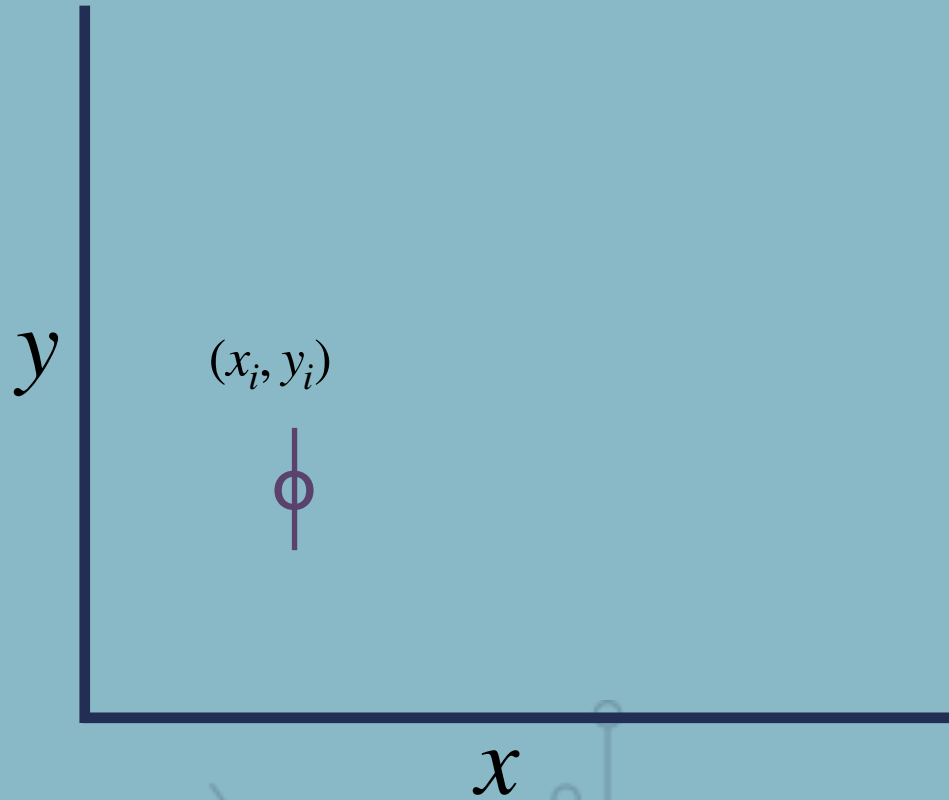
The energy dependence of the pion-proton scattering cross section

$$\sigma(E) = N \frac{\Gamma^2/4}{(E - E_0)^2 + \Gamma^2/4}$$



Goal: estimate E_0

Let's start with a
single data point ...



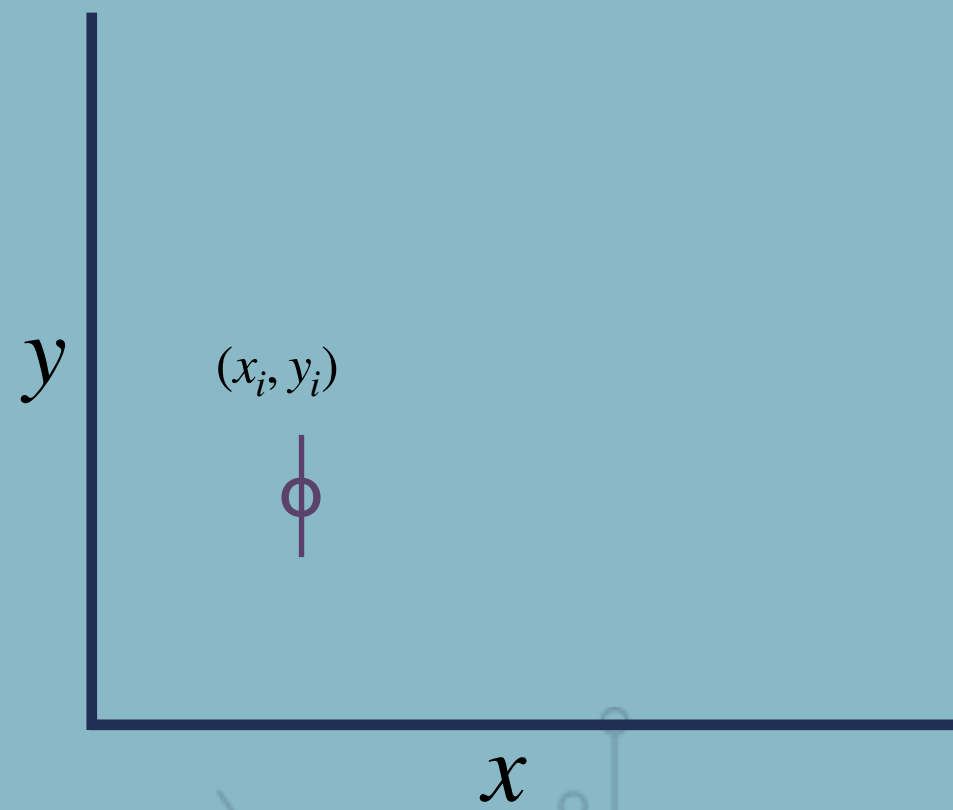
$$y_i$$

$$\sigma_i$$

$$f(x_i, \theta)$$

$$\theta$$

Let's start with a
single data point ...



$$y_i$$

$$\sigma_i$$

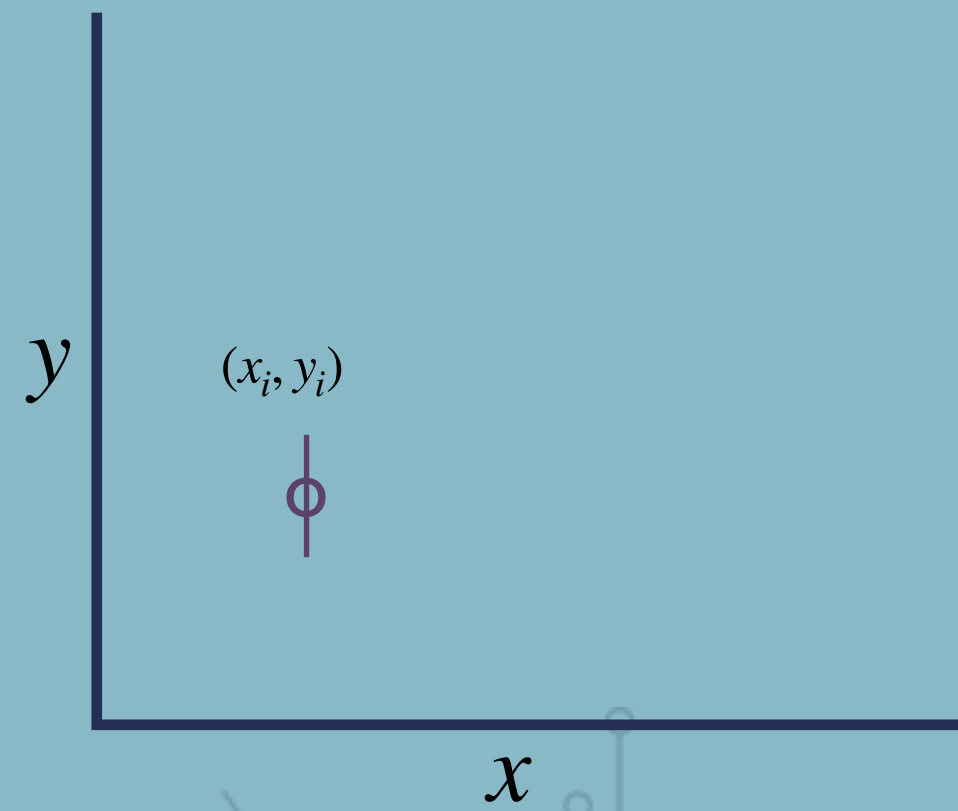
$$f(x_i, \theta)$$

$$\theta$$

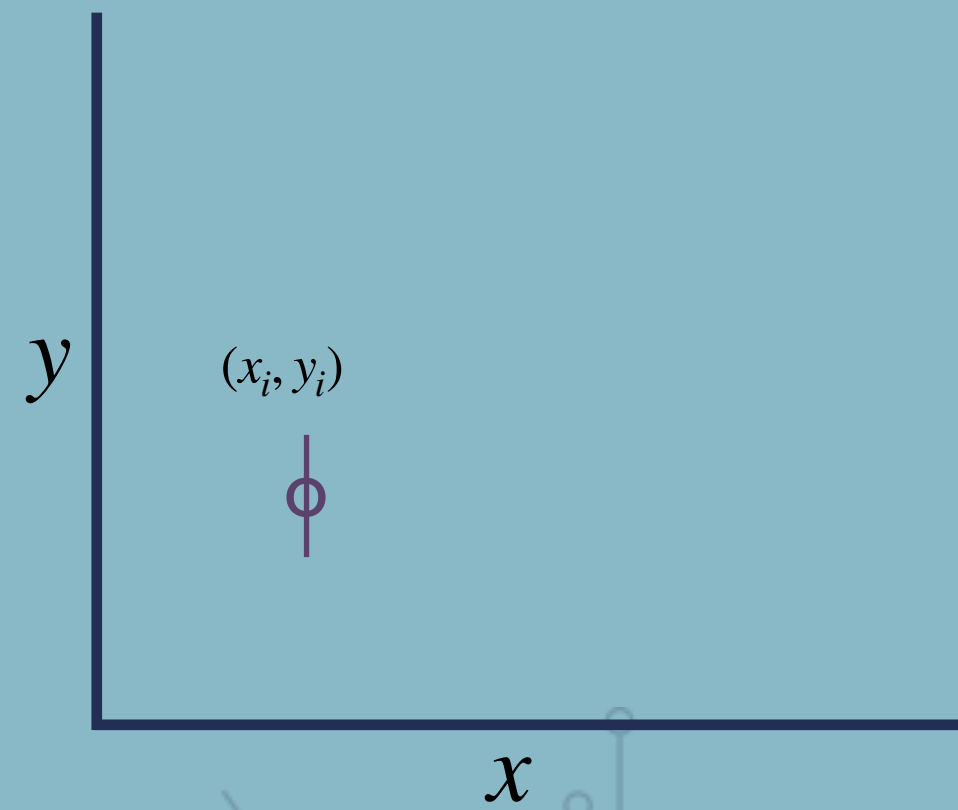
What is the probability of observing the data
given a specific model?

Let's start with a
single data point ...

$$(x_i, y_i, \sigma_i)$$



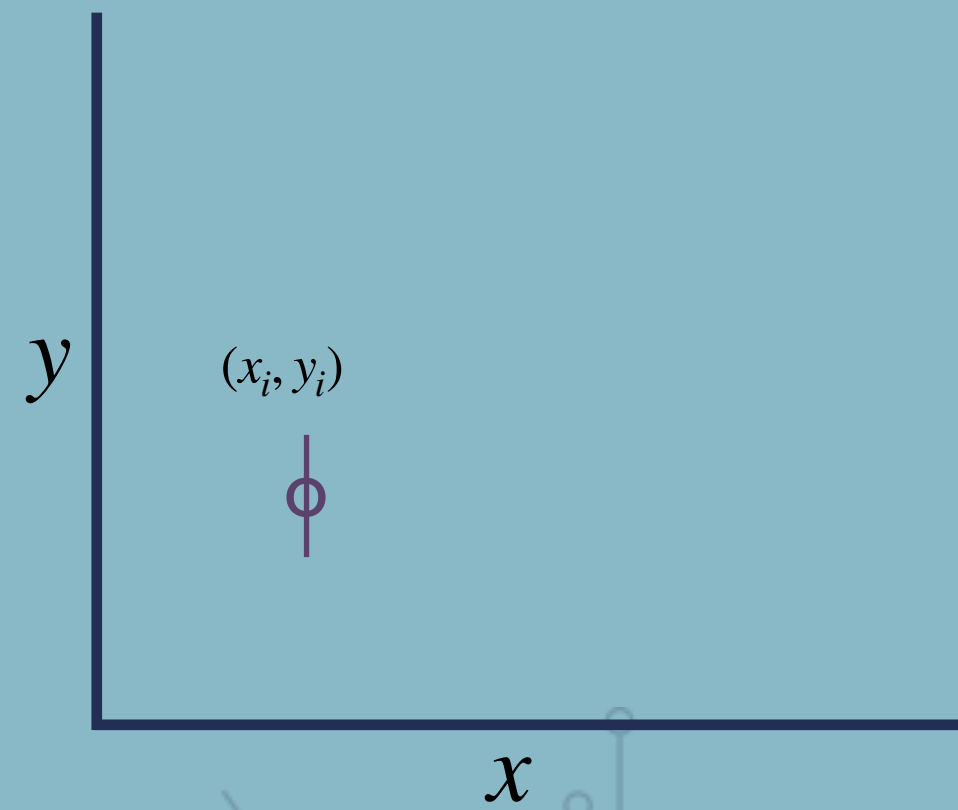
Let's start with a
single data point ...



(x_i, y_i, σ_i)

$$p(y_i | \hat{y}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \hat{y})^2}{2\sigma_i^2}\right)$$

Let's start with a
single data point ...

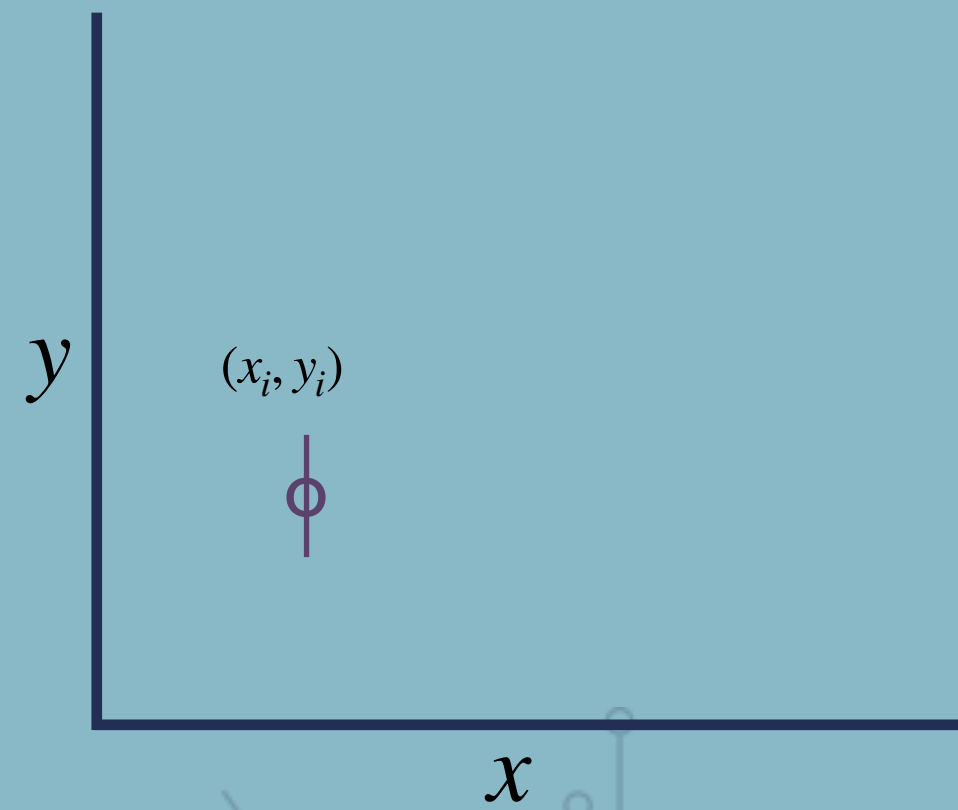


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$$p(y_i | \hat{y}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \hat{y})^2}{2\sigma_i^2}\right)$$

approximate $\hat{y}$ with $f(x_i, \theta)$

Let's start with a single data point ...



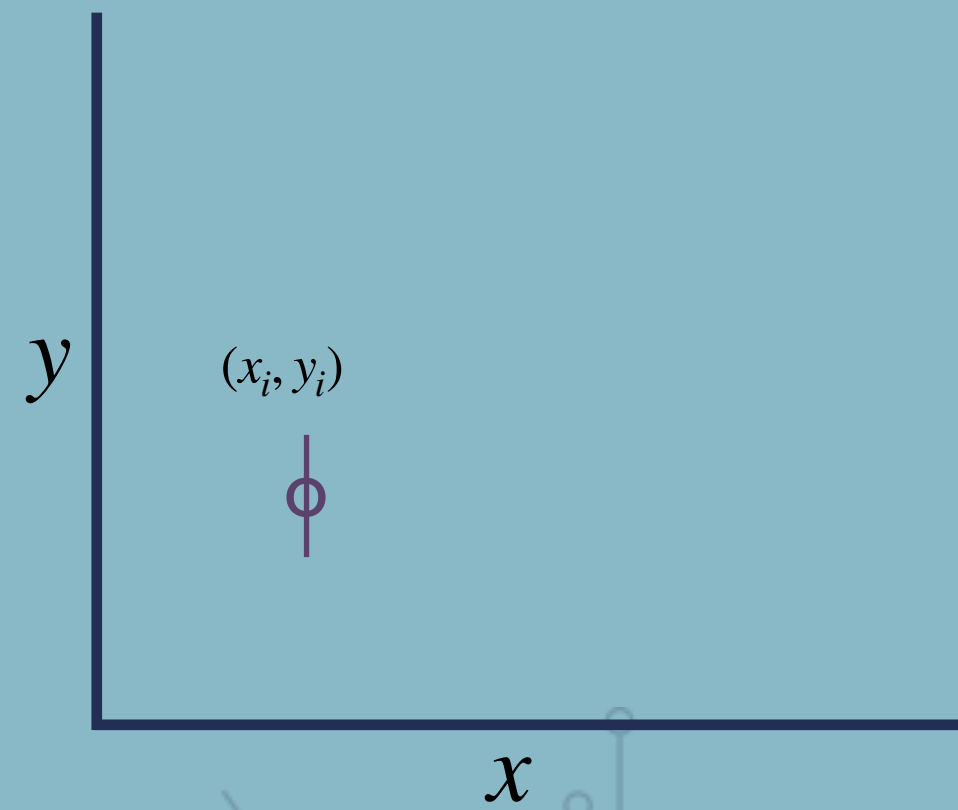
(x_i, y_i, σ_i)

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approximate $\hat{y}$ with $f(x_i, \theta)$

... but which θ should we choose?

Let's start with a single data point ...



(x_i, y_i, σ_i)

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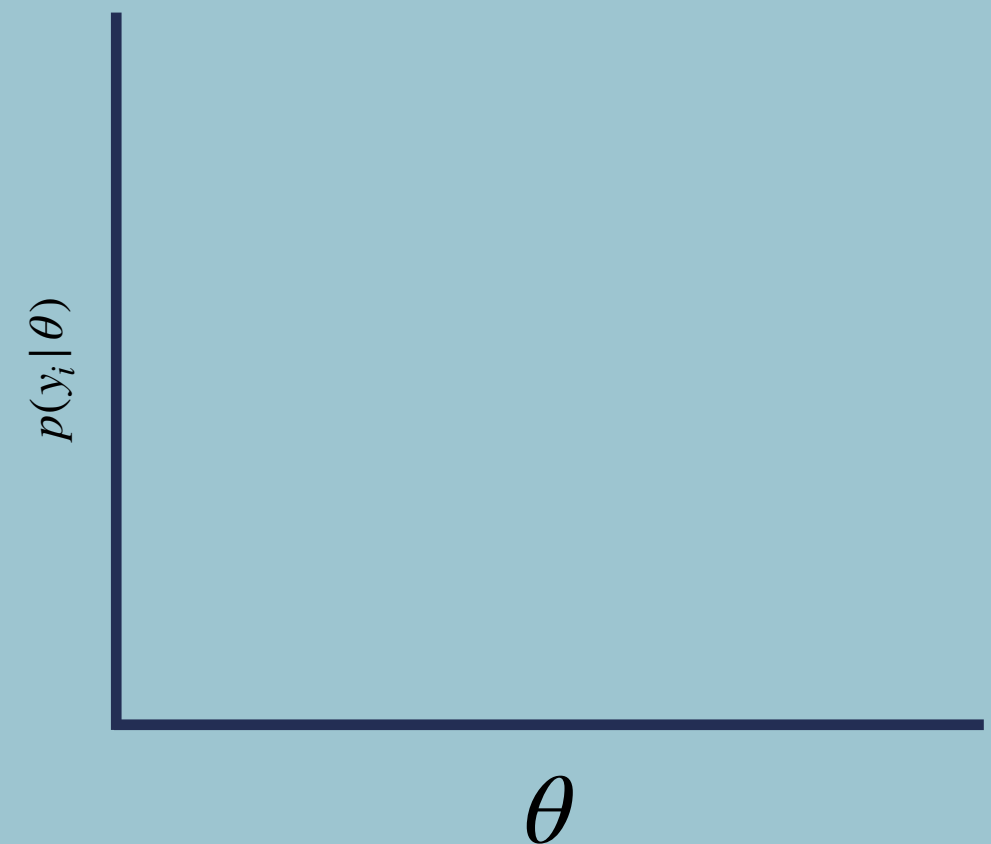
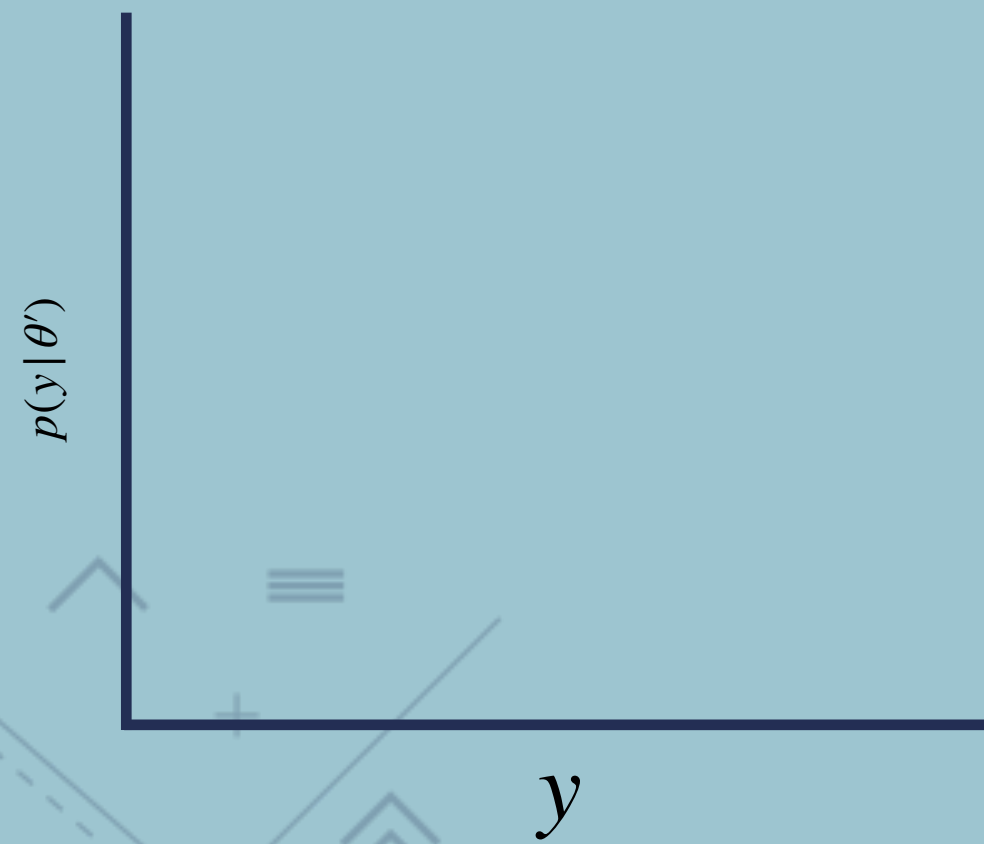
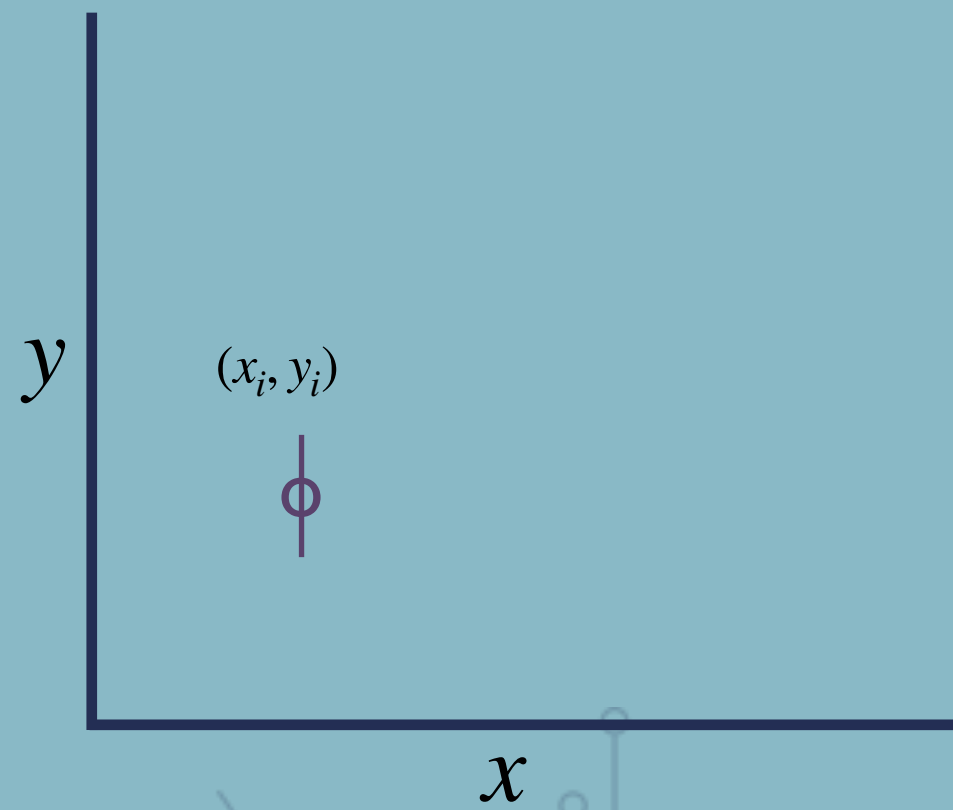
approximate $\hat{y}$ with $f(x_i, \theta)$

... but which θ should we choose?

Idea: if θ is close to the true underlying physical parameter that generated the data, then $f(x_i, \theta)$ should be close to $\hat{y}$

Let's start with a
single data point ...

$$(x_i, y_i, \sigma_i)$$
$$p(y_i | \hat{y}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \hat{y})^2}{2\sigma_i^2}\right)$$

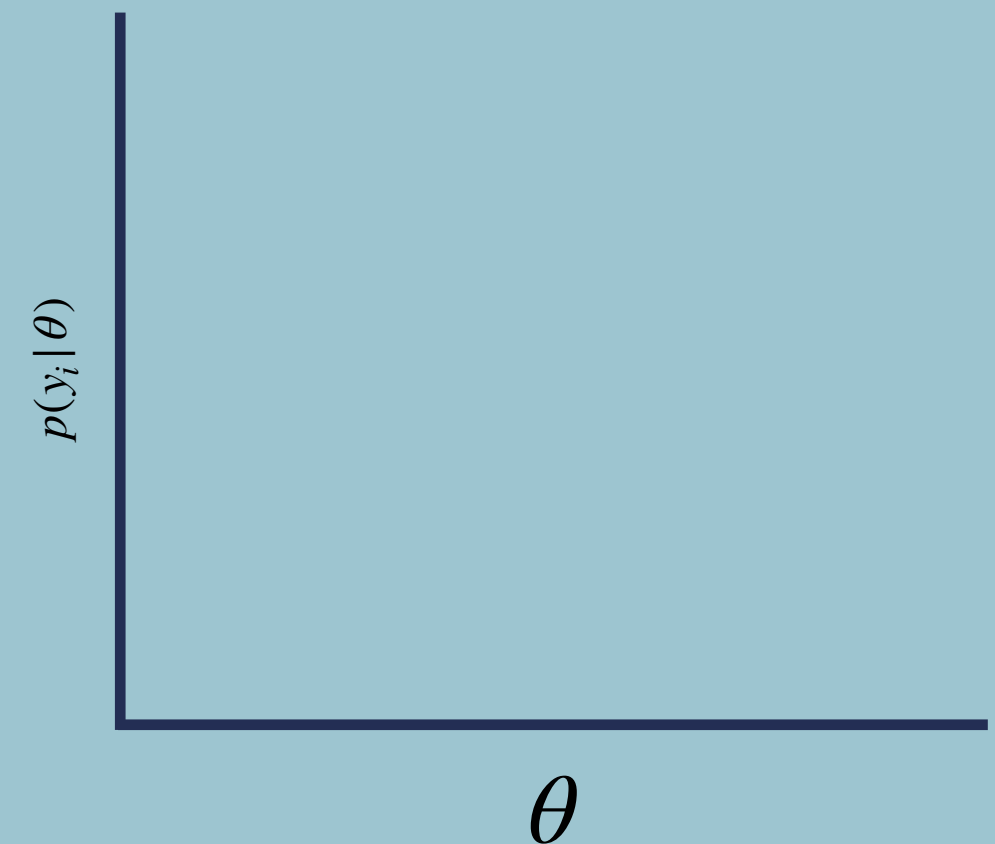
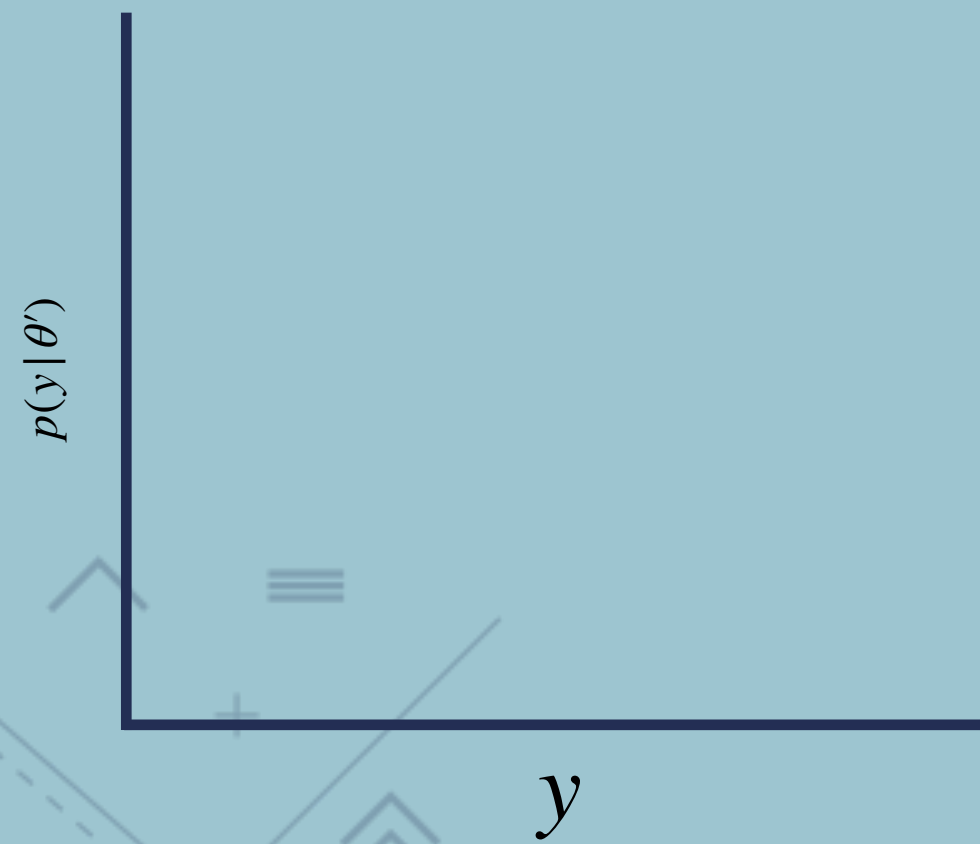
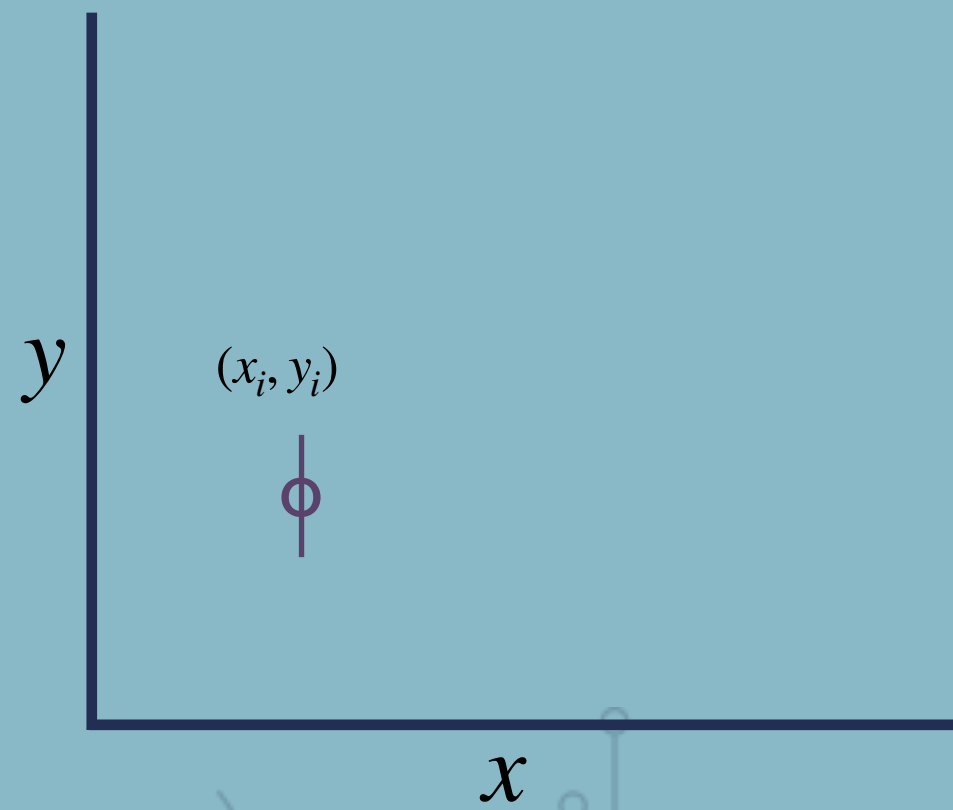


Let's start with a
single data point ...

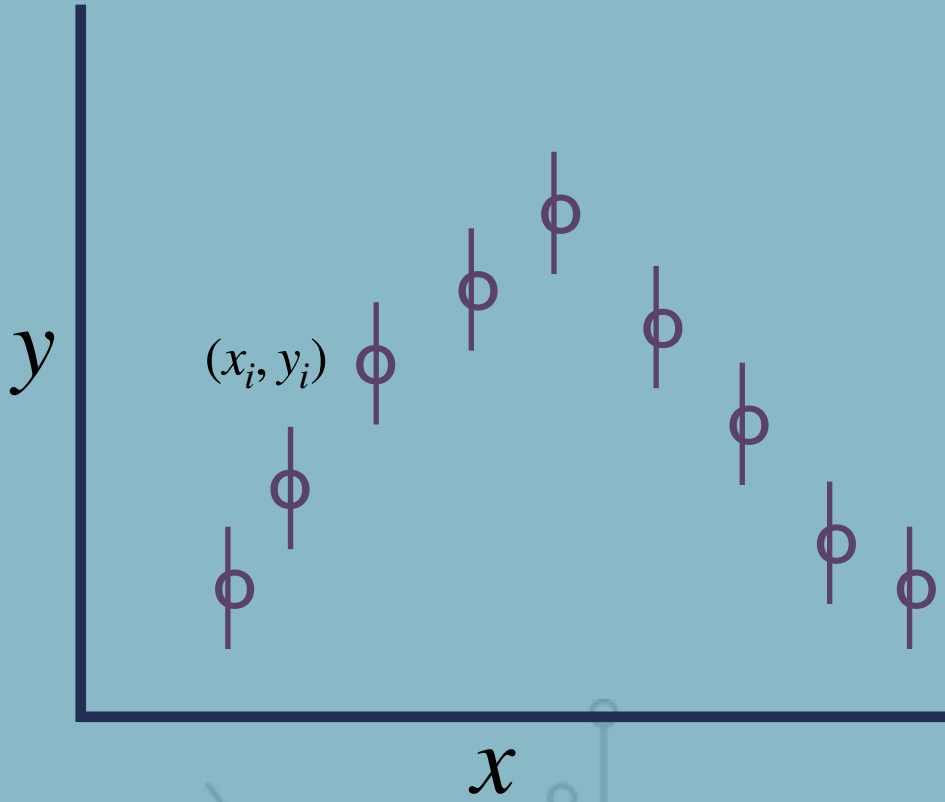
(x_i, y_i, σ_i)

$$p(y_i | \hat{y}) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \hat{y})^2}{2\sigma^2}\right)$$

$$p(y_i | f(x_i, \theta)) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - f(x_i, \theta))^2}{2\sigma^2}\right)$$

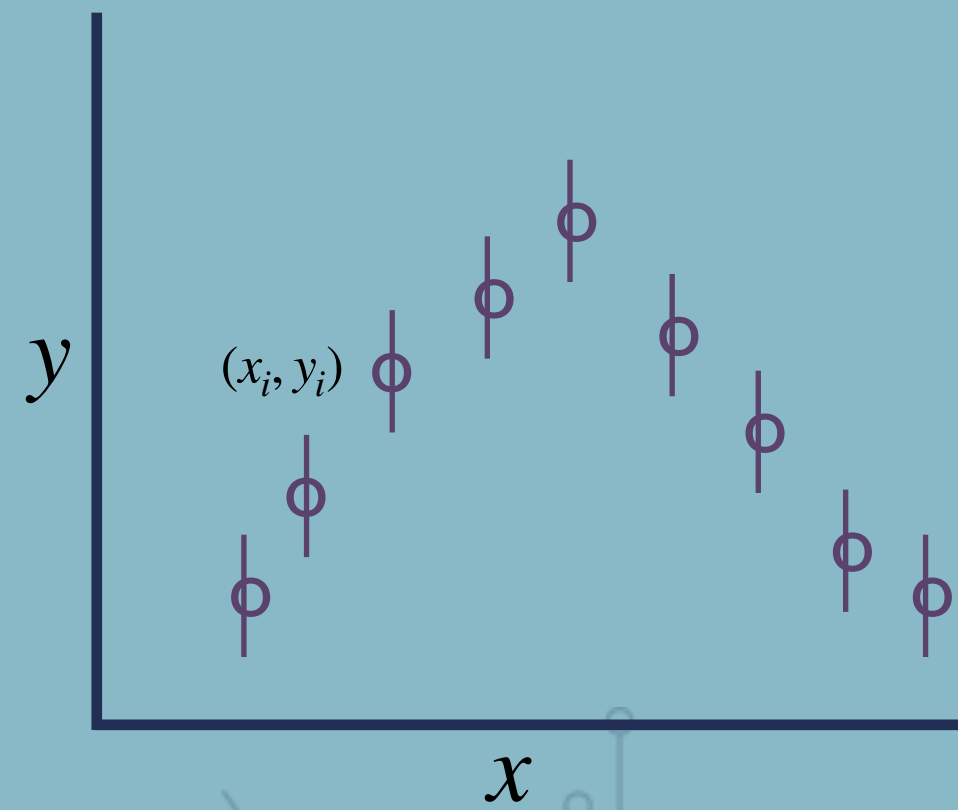


Add more data points



Add more data points

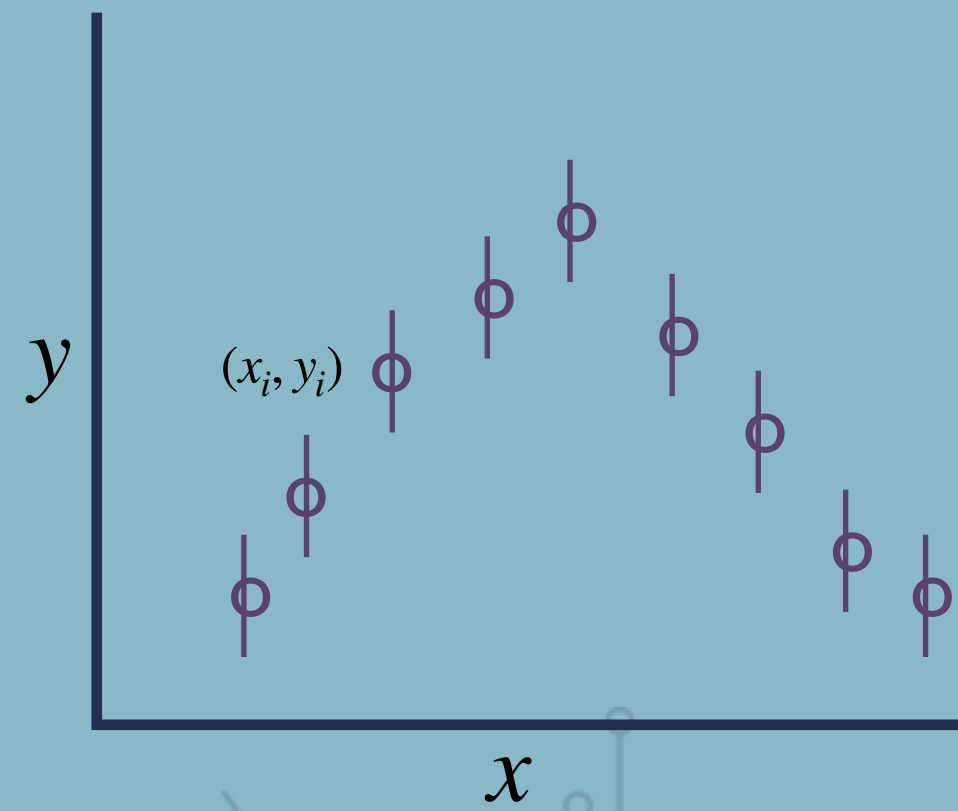
Assume y_i are independent draws from the same distributions



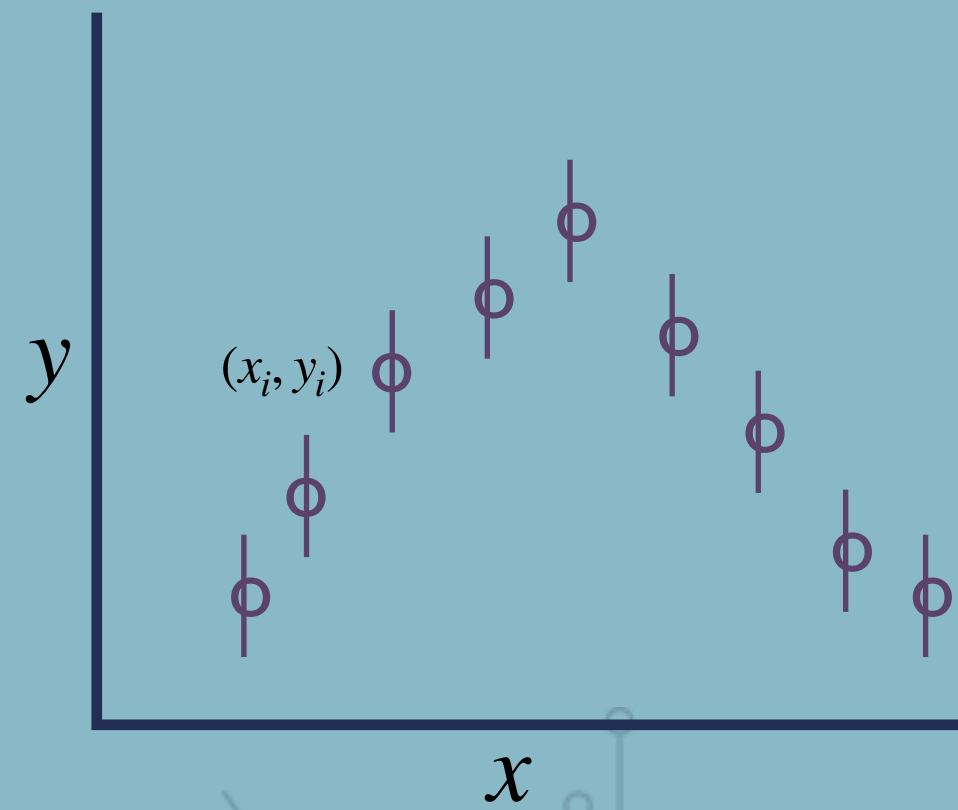
Add more data points

Assume y_i are independent draws from the same distributions

$$y_i \sim \mathcal{N}(f(x_i, \theta), \sigma_i^2)$$



Add more data points



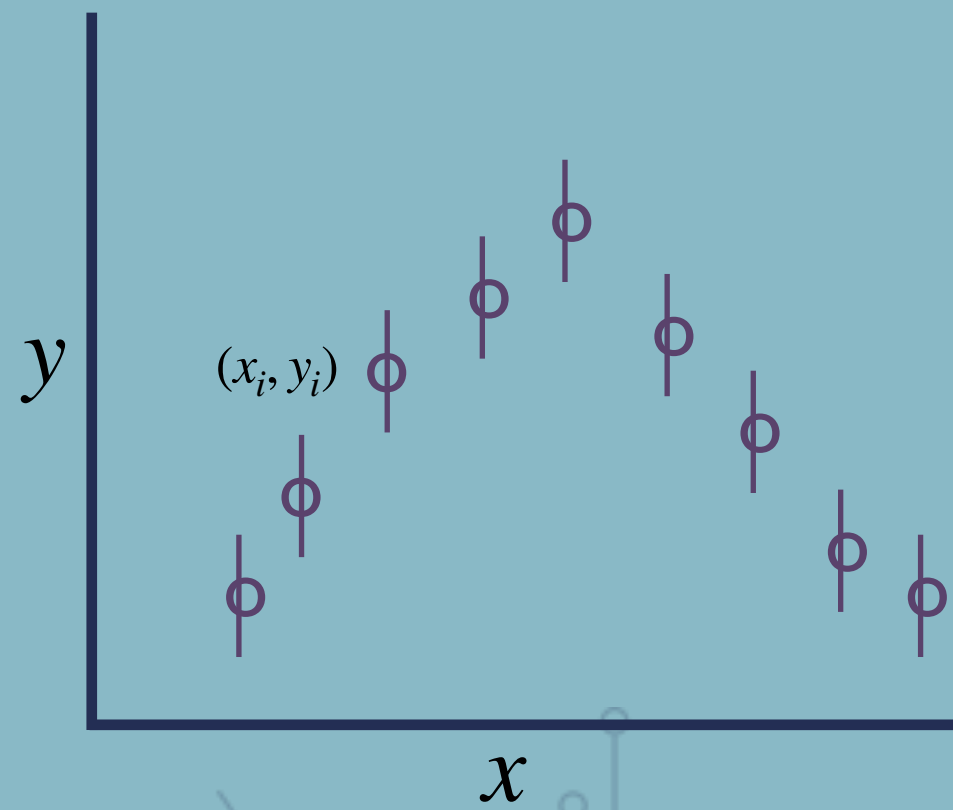
Assume y_i are independent draws from the same distributions

$$y_i \sim \mathcal{N}(f(x_i, \theta), \sigma_i^2)$$

then:

$$p(\bar{y} | \theta) = \prod_{i=1}^N p(y_i | \theta)$$

Add more data points



Assume y_i are independent draws from the same distributions

$$y_i \sim \mathcal{N}(f(x_i, \theta), \sigma_i^2)$$

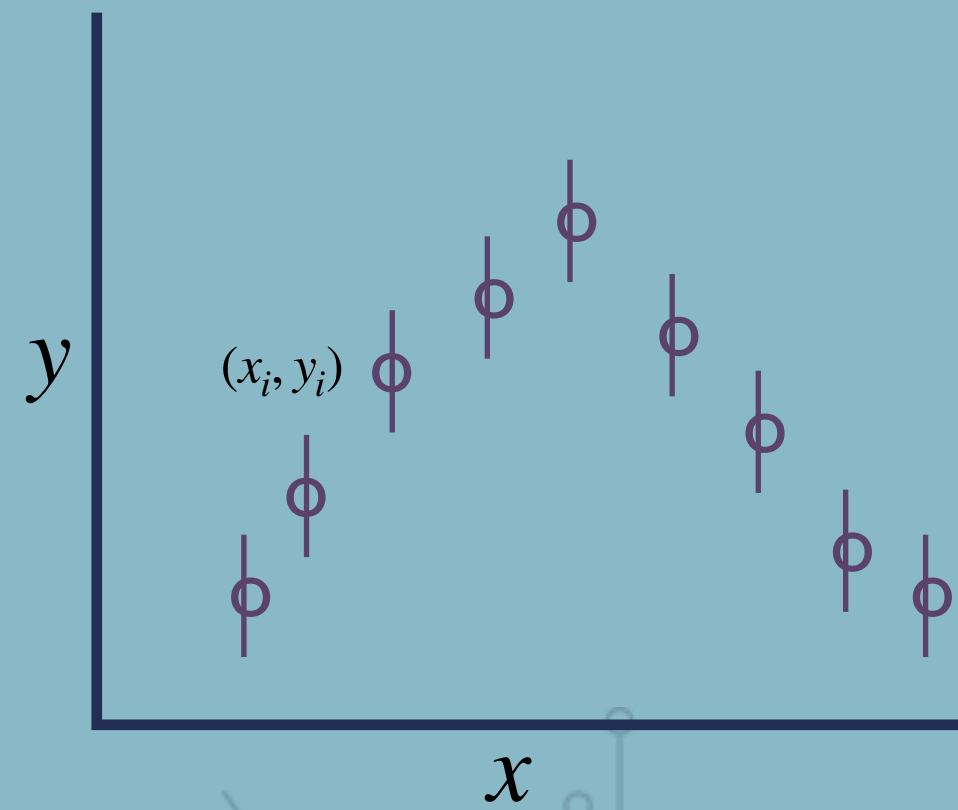
then:

$$p(\bar{y} | \theta) = \prod_{i=1}^N p(y_i | \theta)$$

or in log:

$$\log p(\bar{y} | \theta) = \sum_{i=1}^N \log p(y_i | \theta)$$

Add more data points



Assume y_i are independent draws from the same distributions

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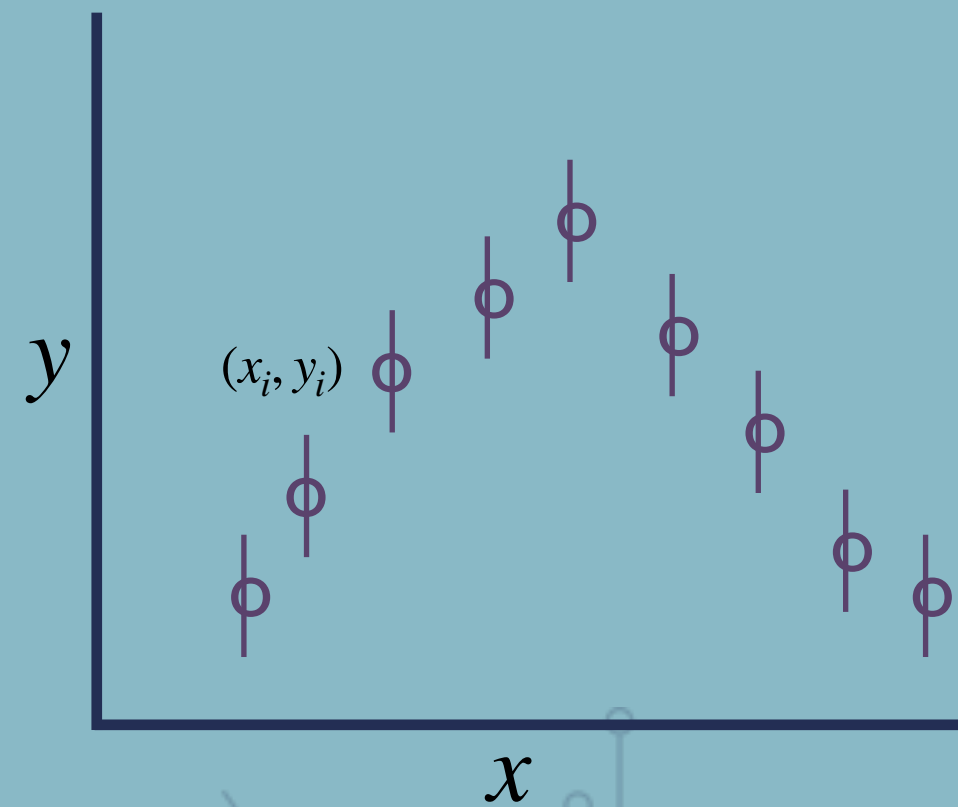
or in log:

$$\log p(\bar{y} | \theta) = \sum_{i=1}^N \log p(y_i | \theta)$$

For our Gaussian distribution:

$$\log p(\bar{y} | \theta) = -\frac{1}{2} \sum_{i=1}^N \log(2\pi\sigma_i^2) - \sum_{i=1}^N \left(\frac{(y_i - f(x_i, \theta))^2}{2\sigma_i^2} \right)$$

Add more data points



Least-squares
regression or χ^2 fitting

Assume y_i are independent draws from the same distributions

$$y_i \sim \mathcal{N}(f(x_i, \theta), \sigma_i^2)$$

then:

$$p(\bar{y} | \theta) = \prod_{i=1}^N p(y_i | \theta)$$

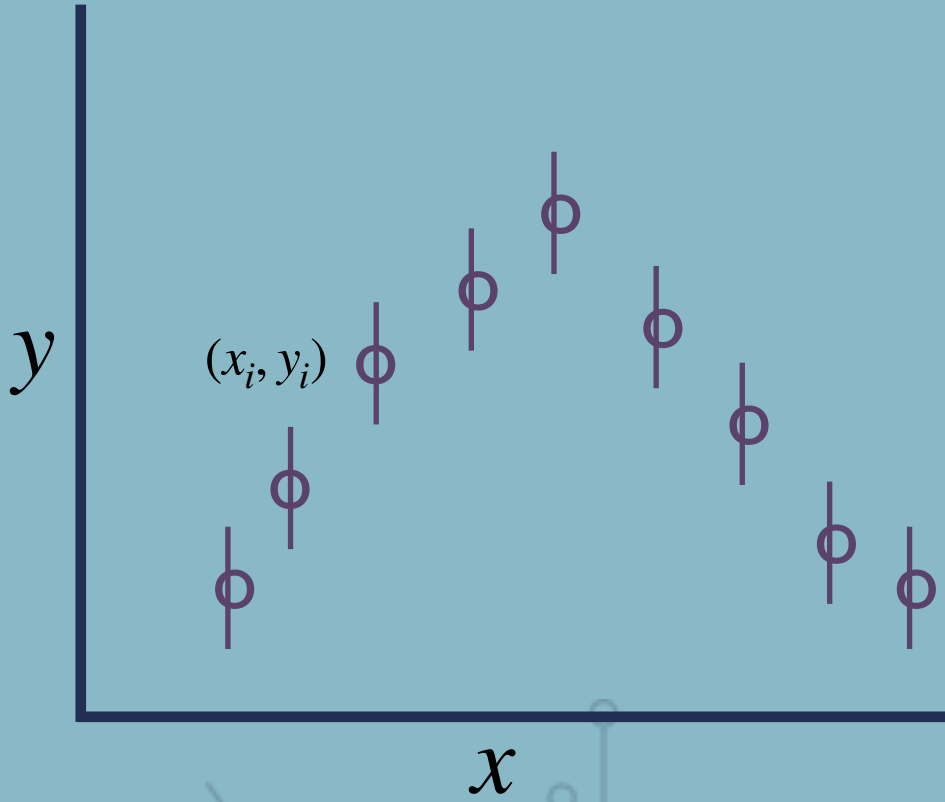
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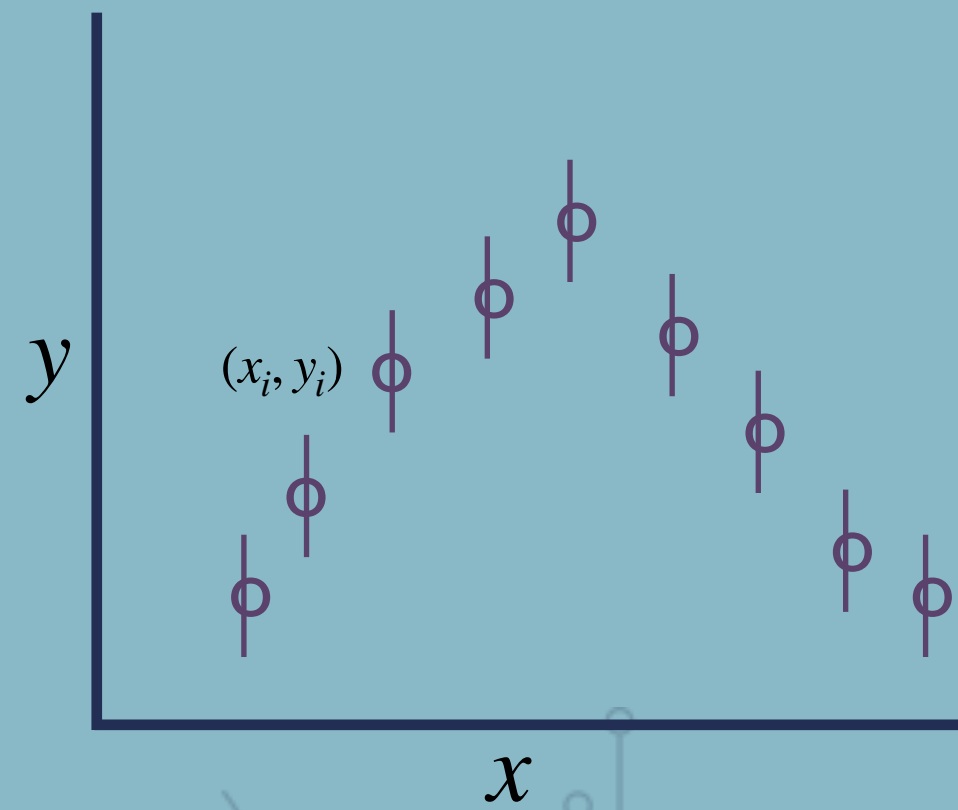
Maximum Likelihood Estimation



Maximum Likelihood Estimation

Find $\hat{\theta}$ where

$$\max_{\theta} p(\bar{y} | \theta)$$

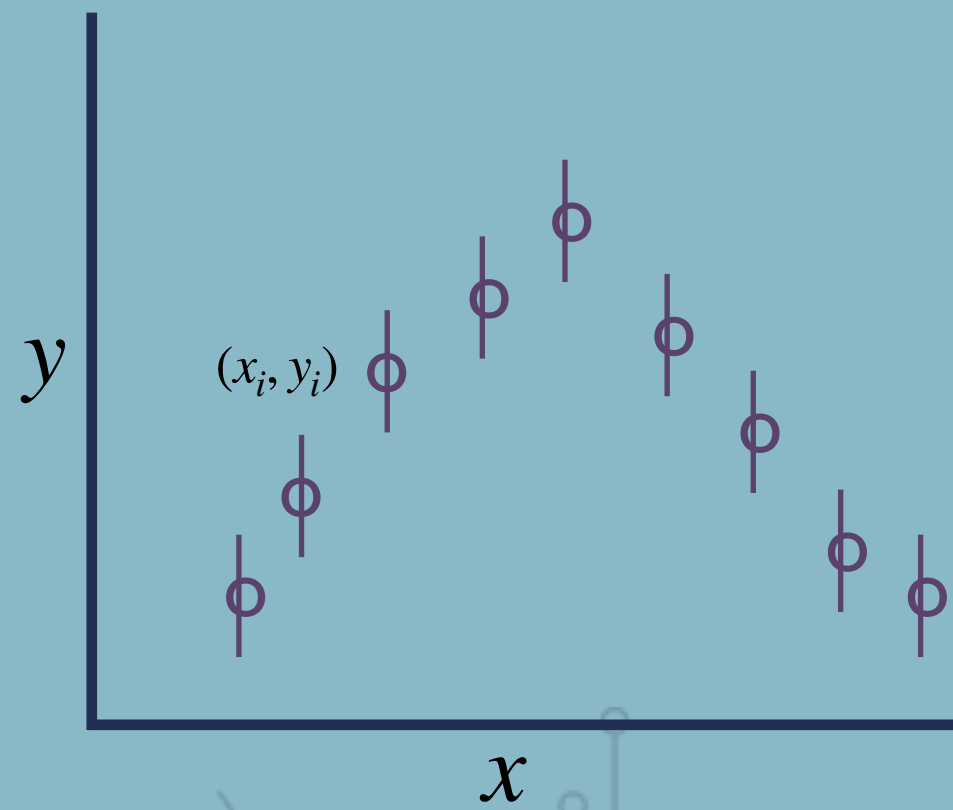


Maximum Likelihood Estimation

Find $\hat{\theta}$ where

$$\max_{\theta} p(\bar{y} | \theta)$$

Note:

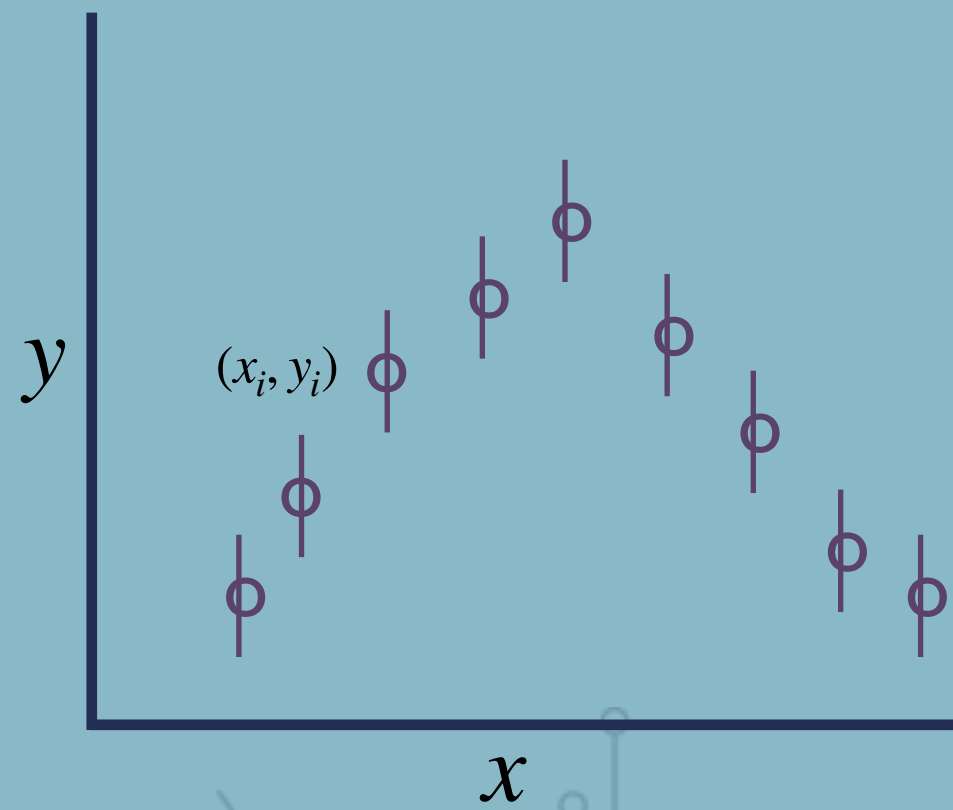


Maximum Likelihood Estimation

Find $\hat{\theta}$ where $\max_{\theta} p(\bar{y} | \theta)$

Note:

- $\max_{\theta} p(\bar{y} | \theta) = \max_{\theta} \log p(\bar{y} | \theta)$

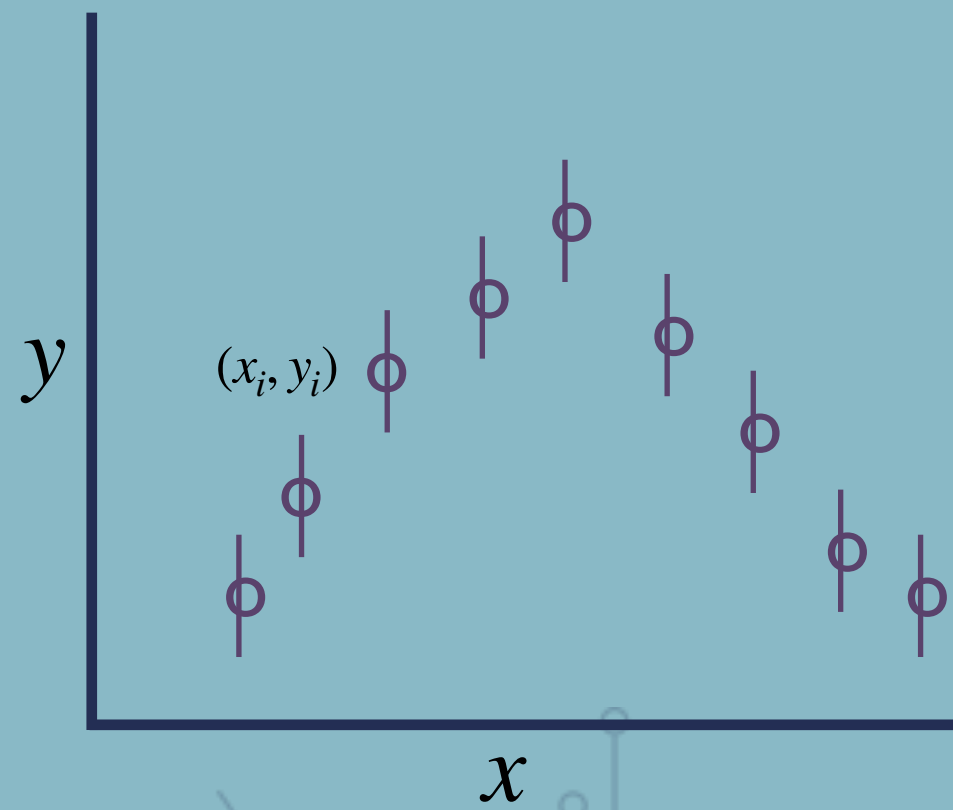


Maximum Likelihood Estimation

Find $\hat{\theta}$ where $\max_{\theta} p(\bar{y} | \theta)$

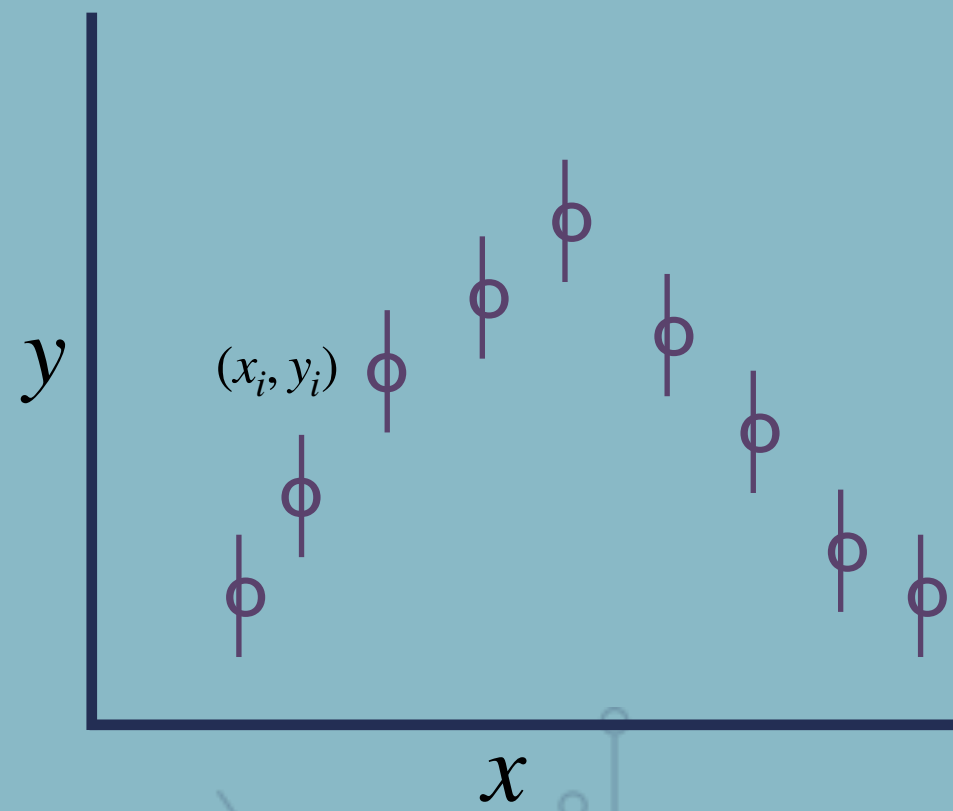
Note:

- $\max_{\theta} p(\bar{y} | \theta) = \max_{\theta} \log p(\bar{y} | \theta)$
- $\max_{\theta} p(\bar{y} | \theta) = \min_{\theta} - \log p(\bar{y} | \theta)$



More Formally

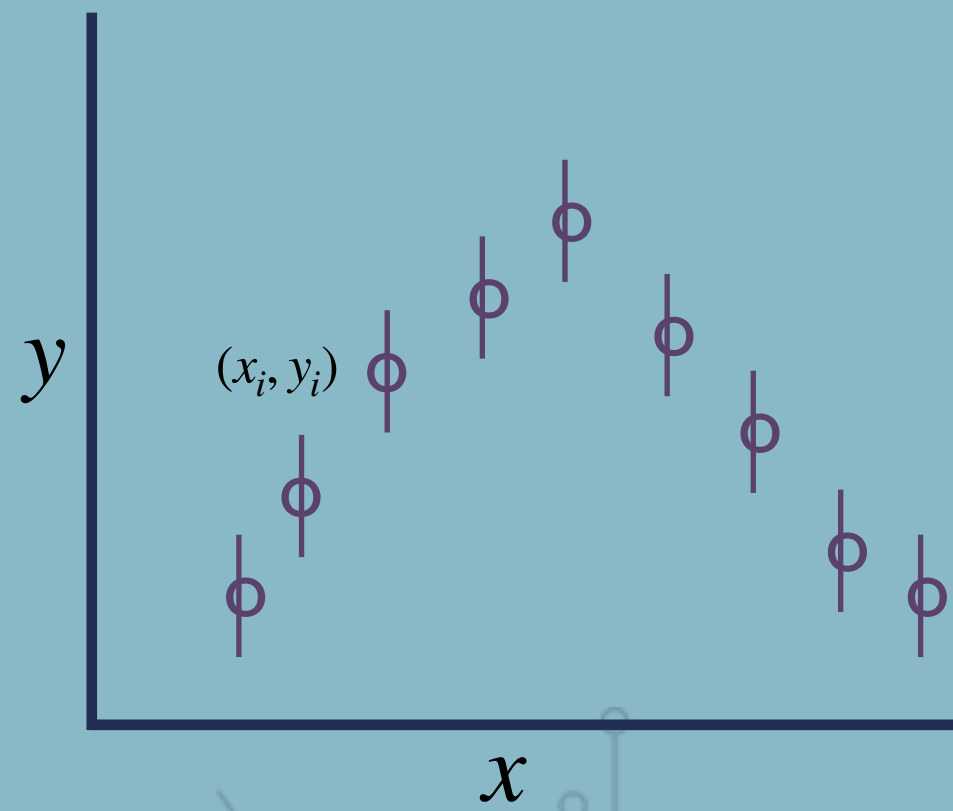
$$E[Y|X] = f(X, \theta) + \epsilon$$



More Formally

Estimate functional relationships

$$E[Y|X] = f(X, \theta) + \epsilon$$

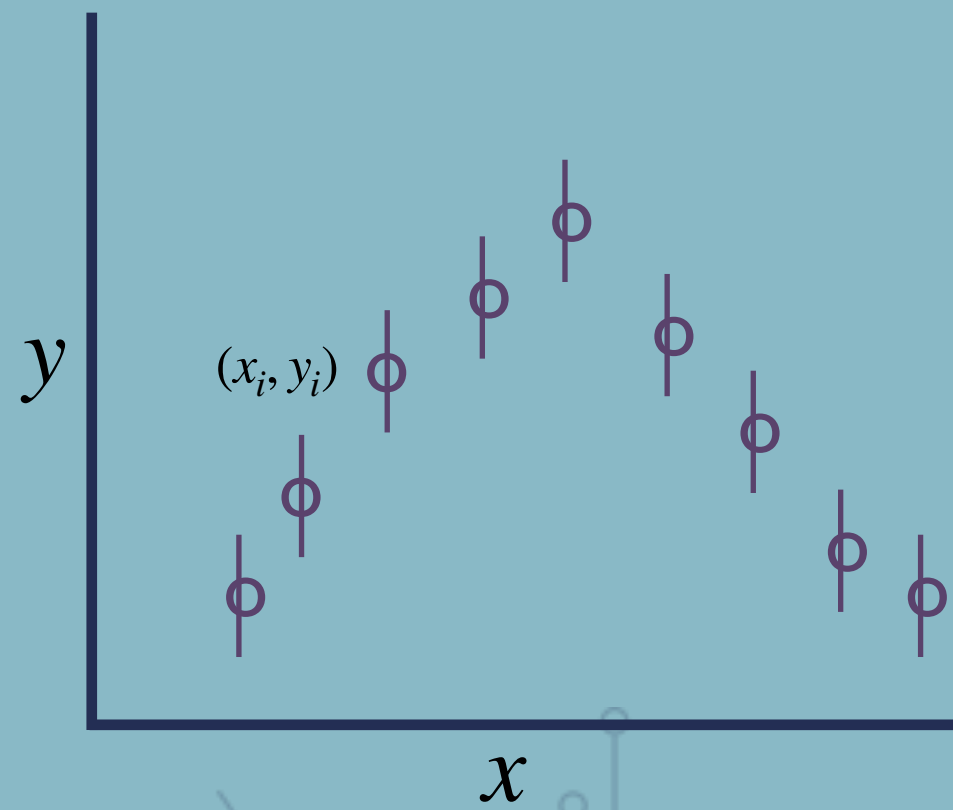


More Formally

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$$E[Y|X] = f(X, \theta) + \epsilon$$

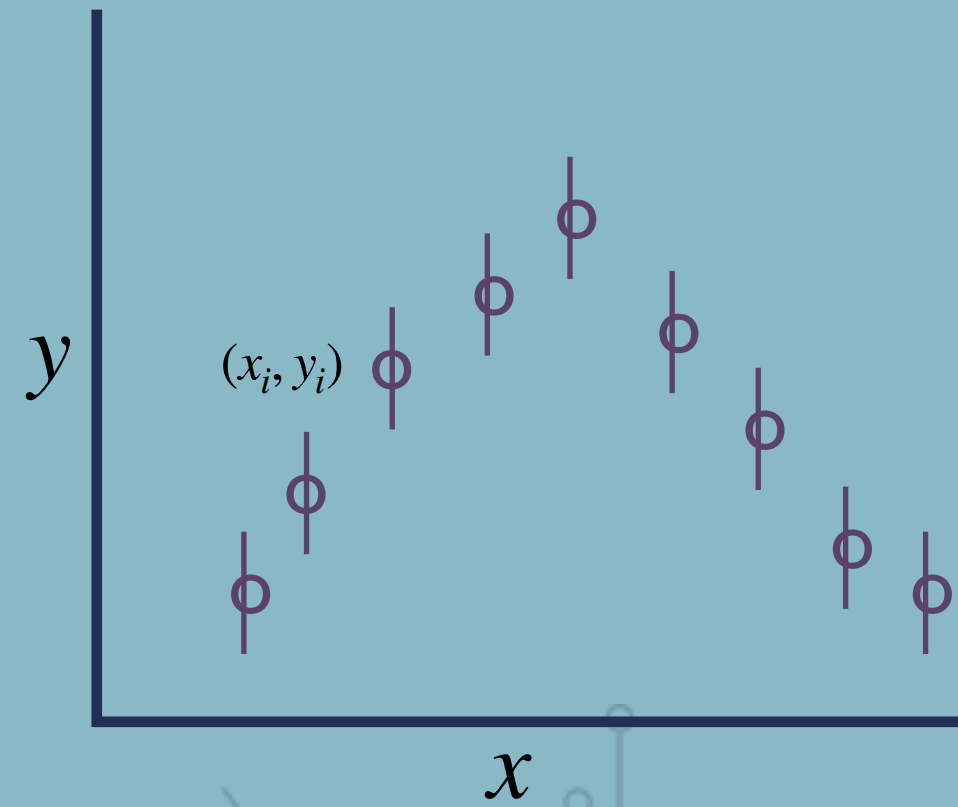
- Homoscedastic errors: ϵ is a n -vector with σ^2



More Formally

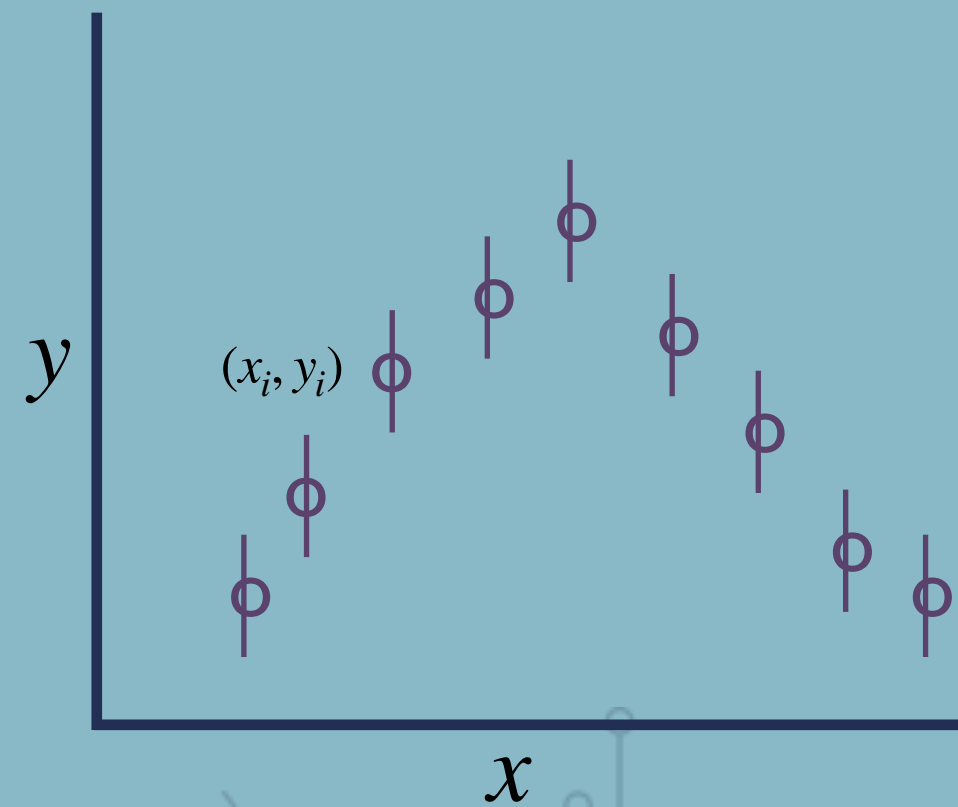
Estimate functional relationships

$$E[Y | X] = f(X, \theta) + \epsilon$$



- Homoscedastic errors: ϵ is a n -vector with σ^2
- Heteroscedastic errors: ϵ is a n -vector with σ_i^2

More Formally

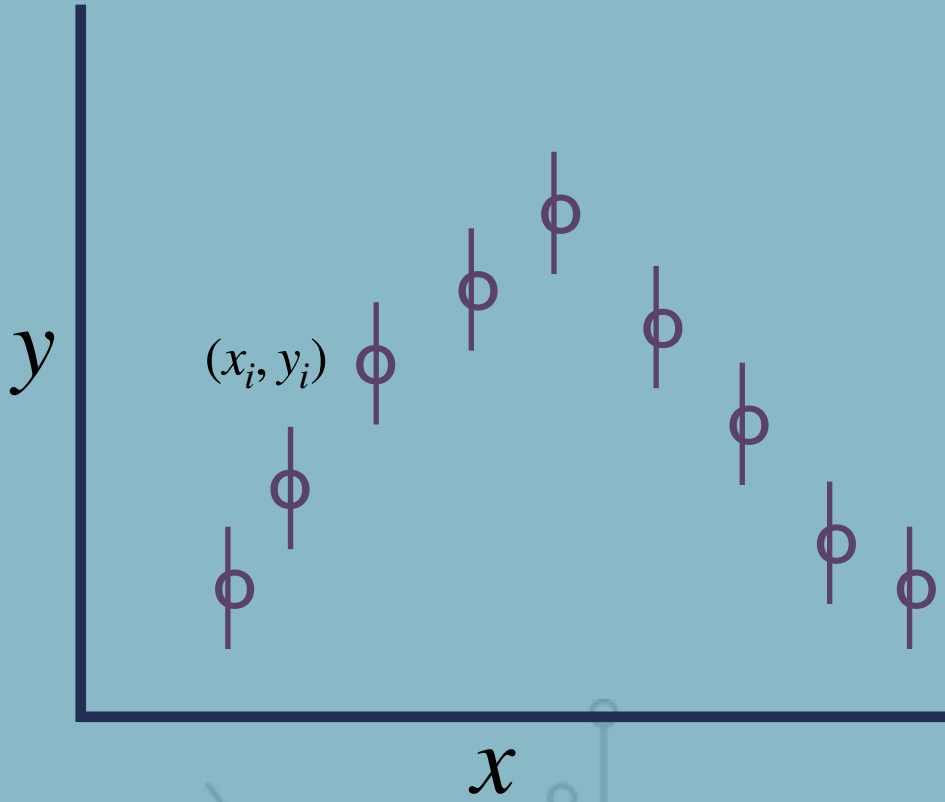


Estimate functional relationships

$$E[Y | X] = f(X, \theta) + \epsilon$$

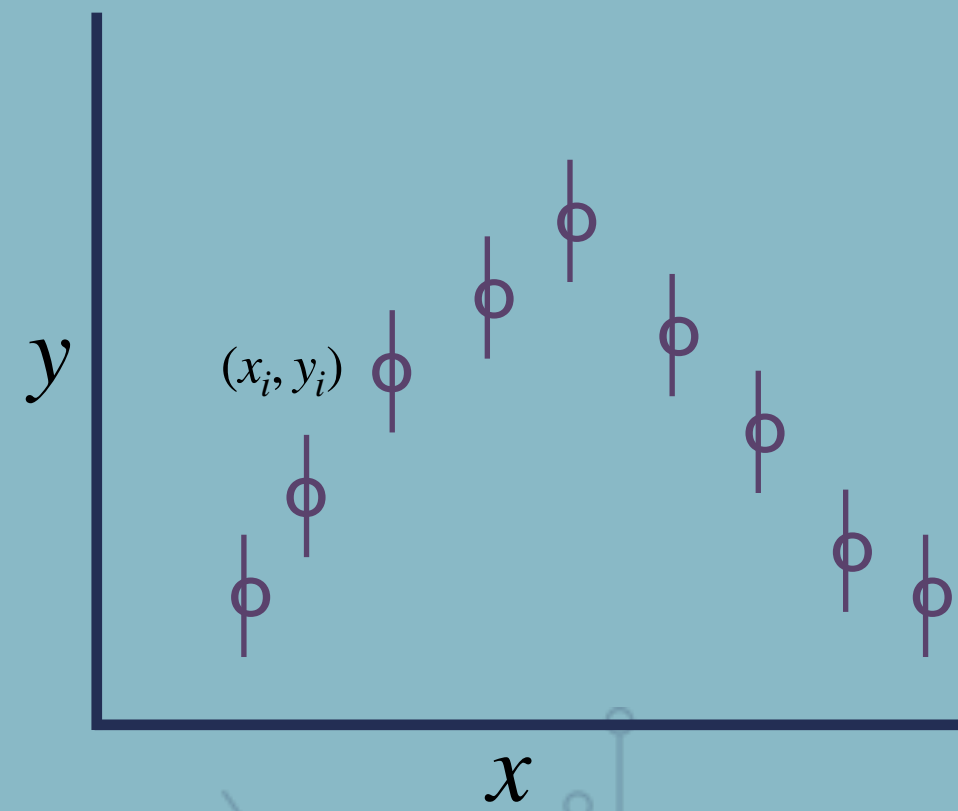
- Homoscedastic errors: ϵ is a n -vector with σ^2
- Heteroscedastic errors: ϵ is a n -vector with σ_i^2
- Errors-in-variables models assume X has error as well

More Formally



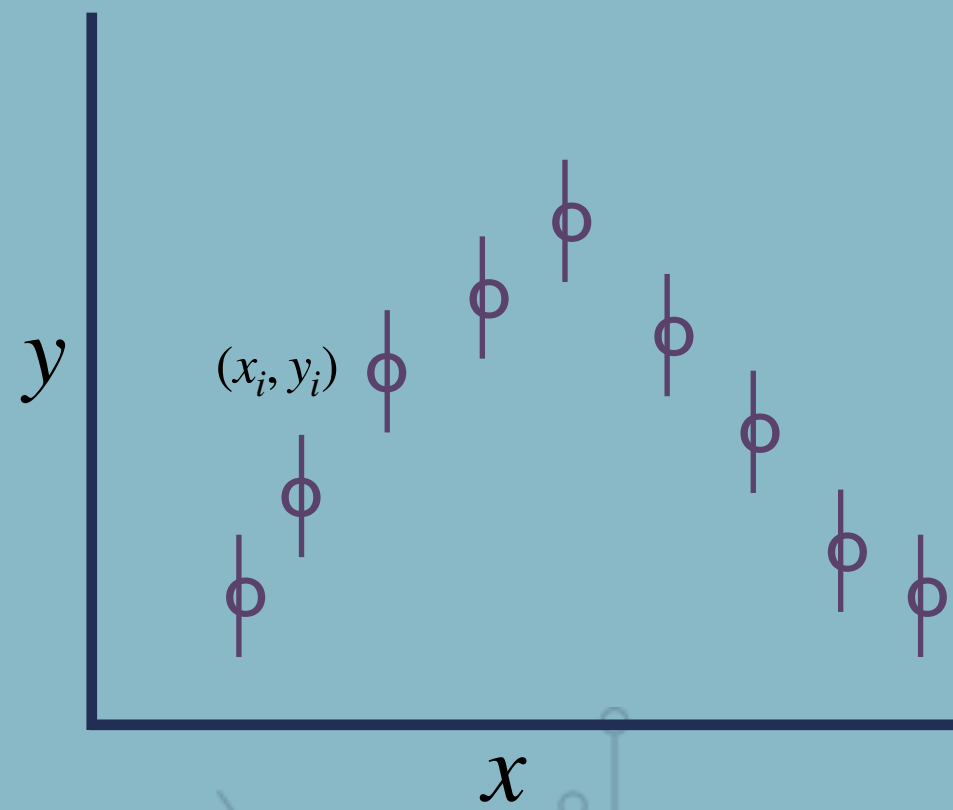
More Formally

- In statistics, a **model** describes

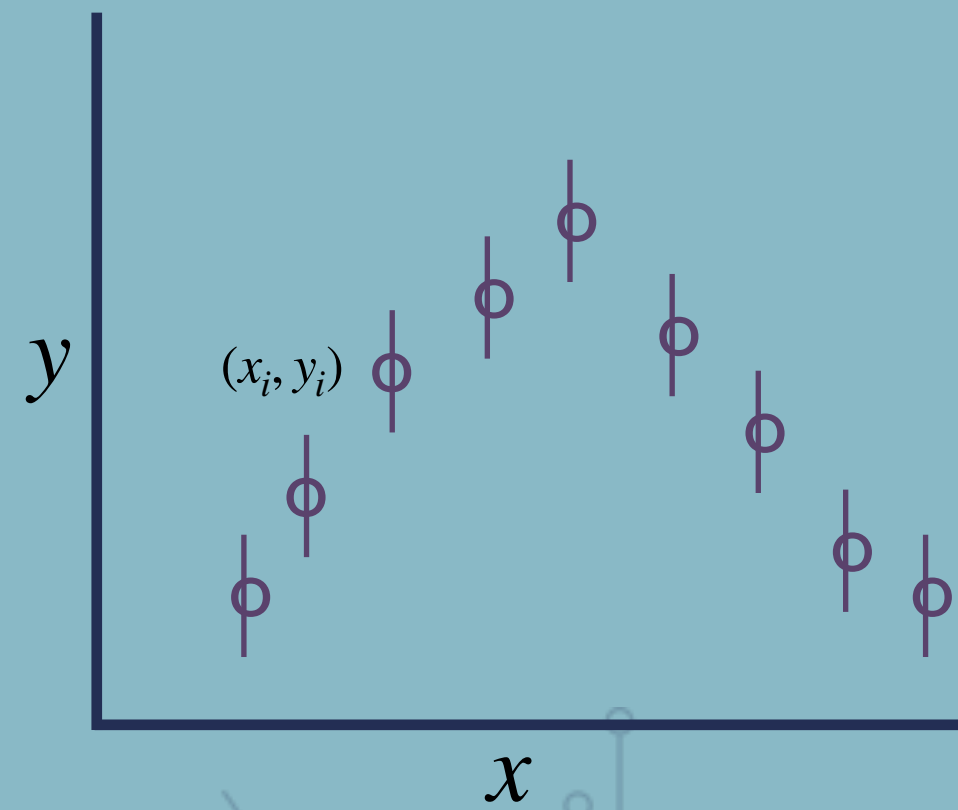


More Formally

- In statistics, a **model** describes
 - the **(functional) relationship** between explanatory variables (which we control) and response variables (data points),

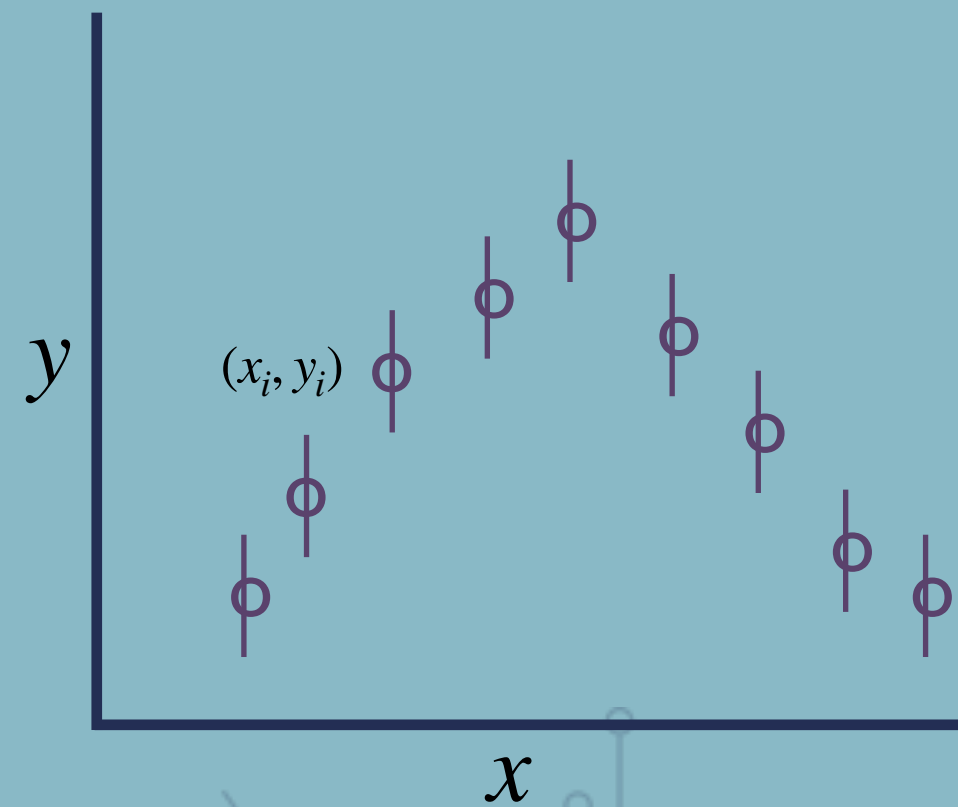


More Formally



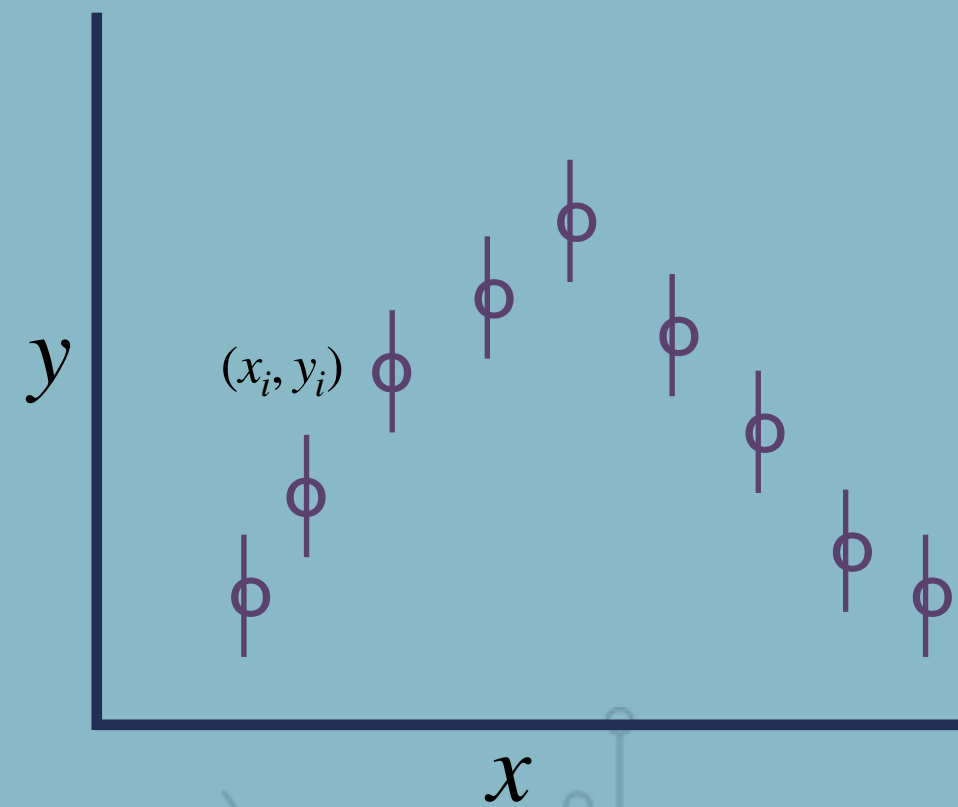
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 - The **statistical assumptions** about data uncertainties and so on (e.g. the likelihood, priors, ...)

More Formally



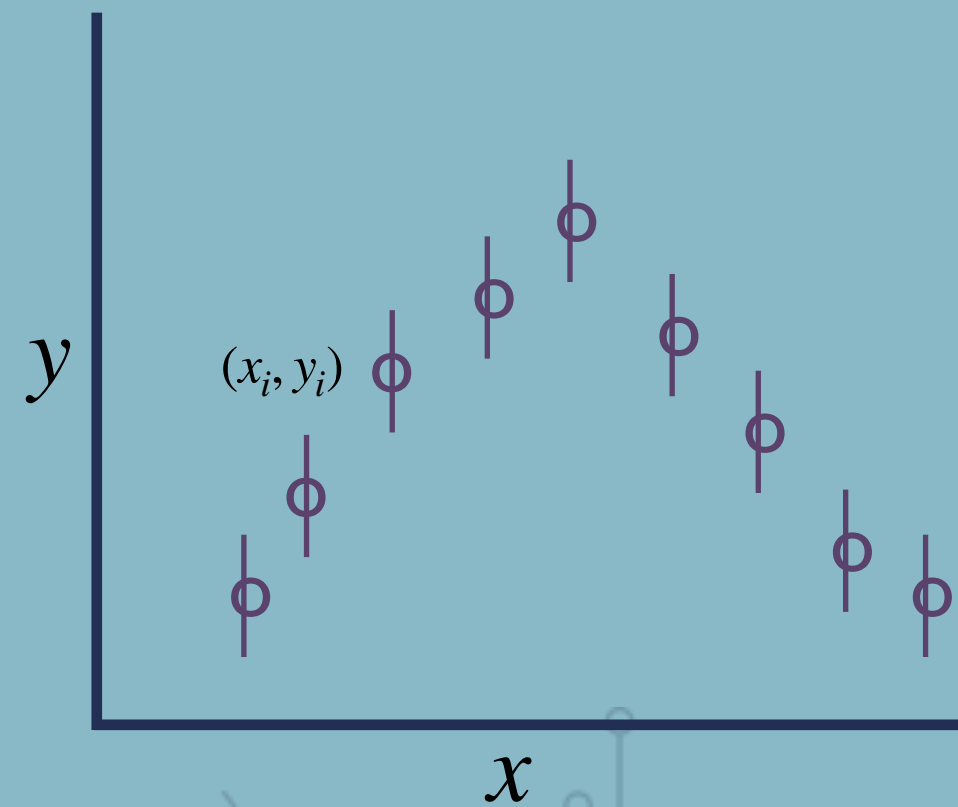
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 - The choice of model and all other underlying assumptions (e.g. decisions made during data pre-processing)

More Formally



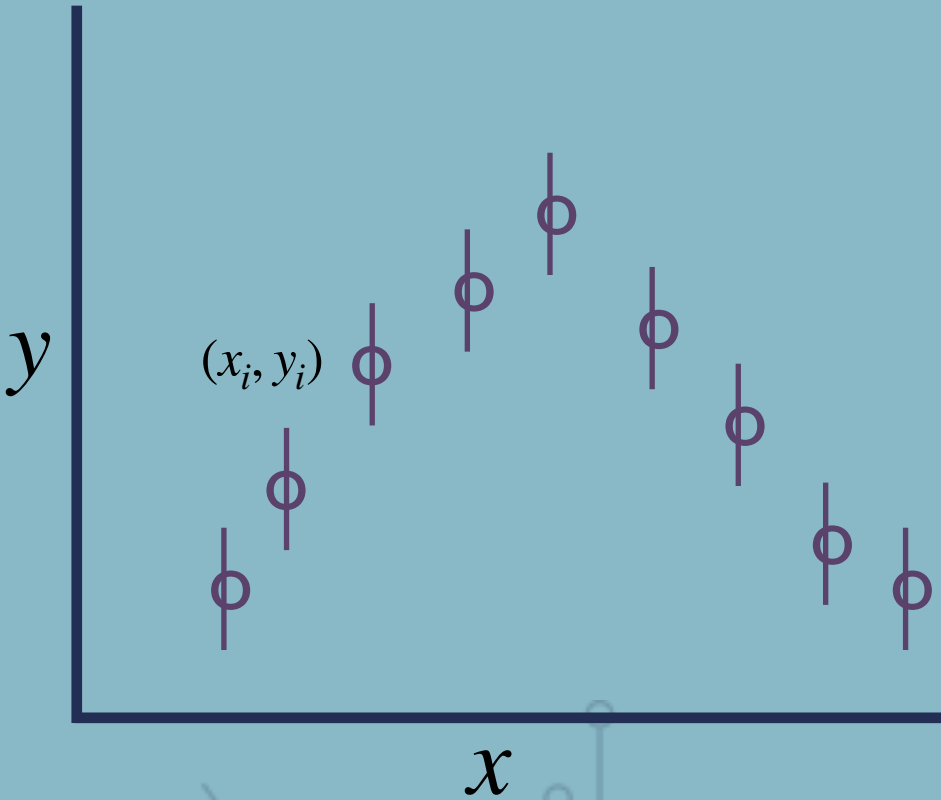
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 - The choice of model and all other underlying assumptions (e.g. decisions made during data pre-processing)
- Simple model: no unknown terms

More Formally



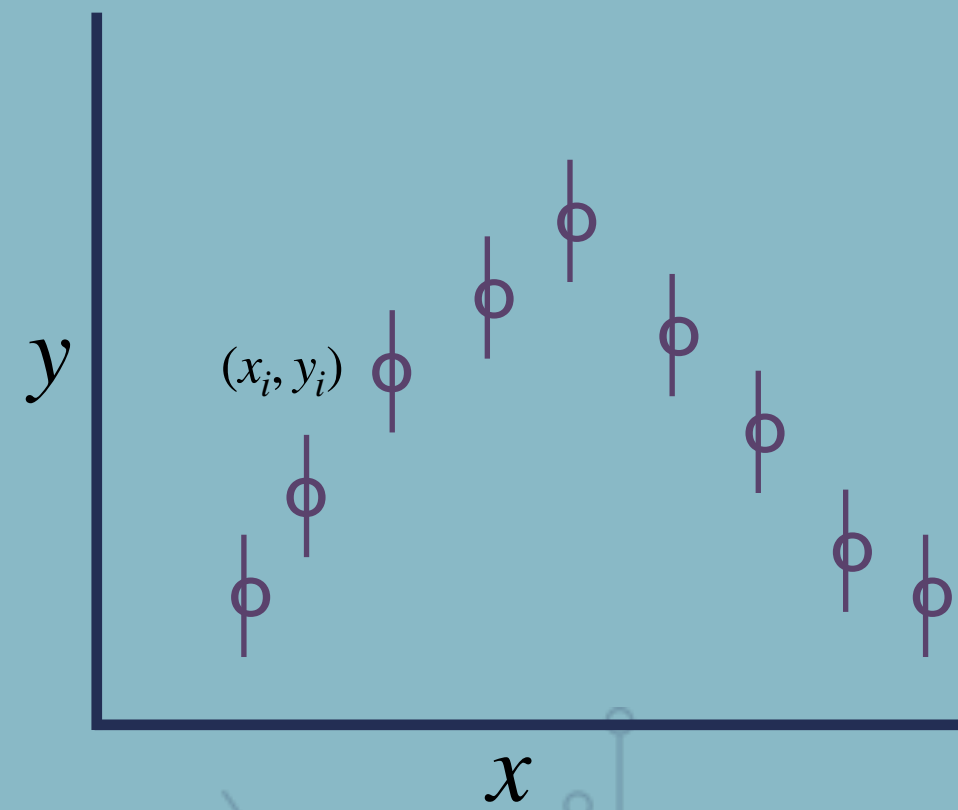
- In statistics, a **model** describes
 - the **(functional) relationship** between explanatory variables (which we control) and response variables (data points),
 - The **statistical assumptions** about data uncertainties and so on (e.g. the likelihood, priors, ...)
 - The choice of model and all other underlying assumptions (e.g. decisions made during data pre-processing)
- Simple model: no unknown terms
- Composite hypothesis: one or more unknown terms = the model's free parameters

In Practice

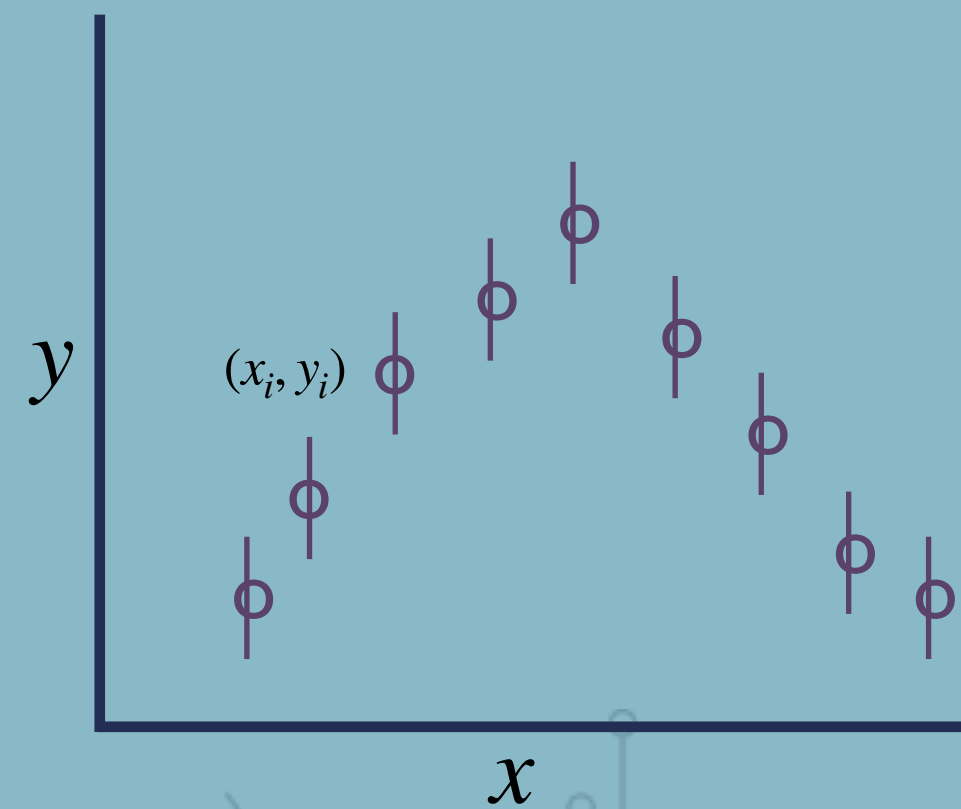


In Practice

- The likelihood spans a function in the space spanned by all the free parameters (e.g. for 3 free parameters, the space is 3-dimensional)

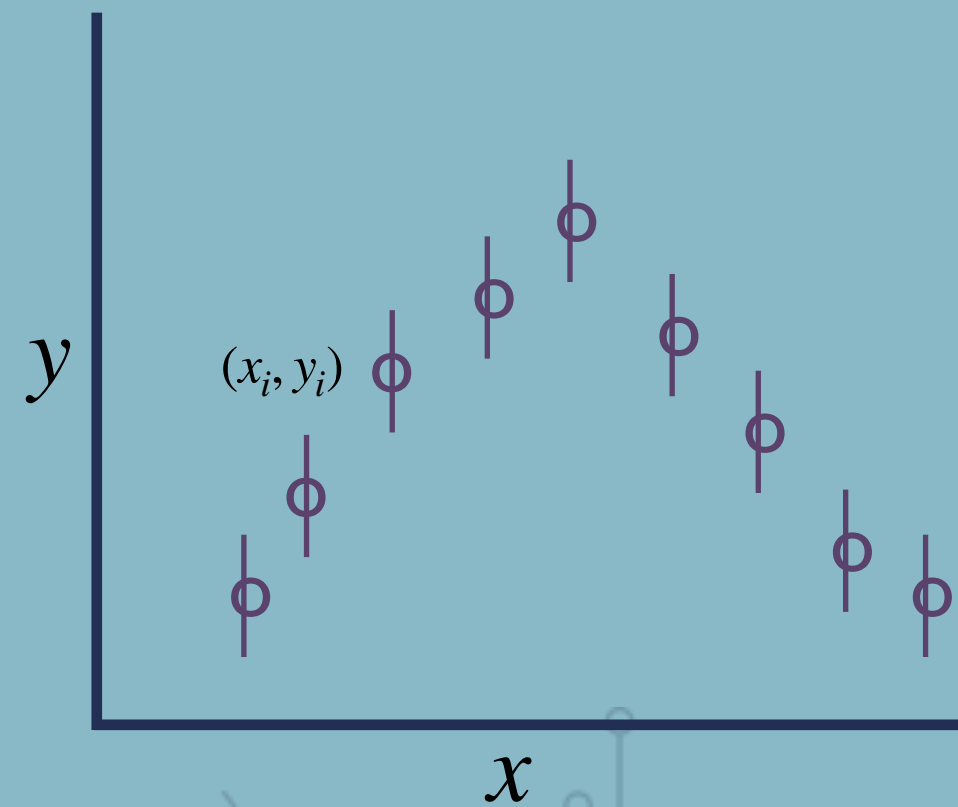


In Practice



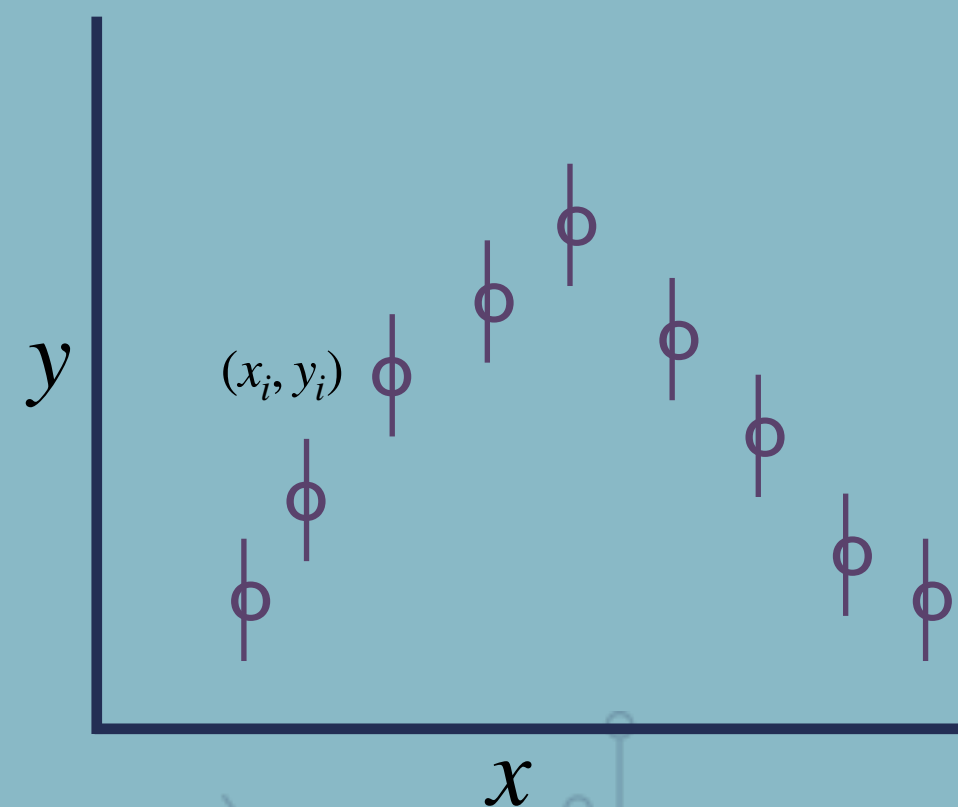
- The likelihood spans a function in the space spanned by all the free parameters (e.g. for 3 free parameters, the space is 3-dimensional)
- To find the maximum likelihood (“best fit”) of your model parameters, find the maximum of that function in that space:

In Practice



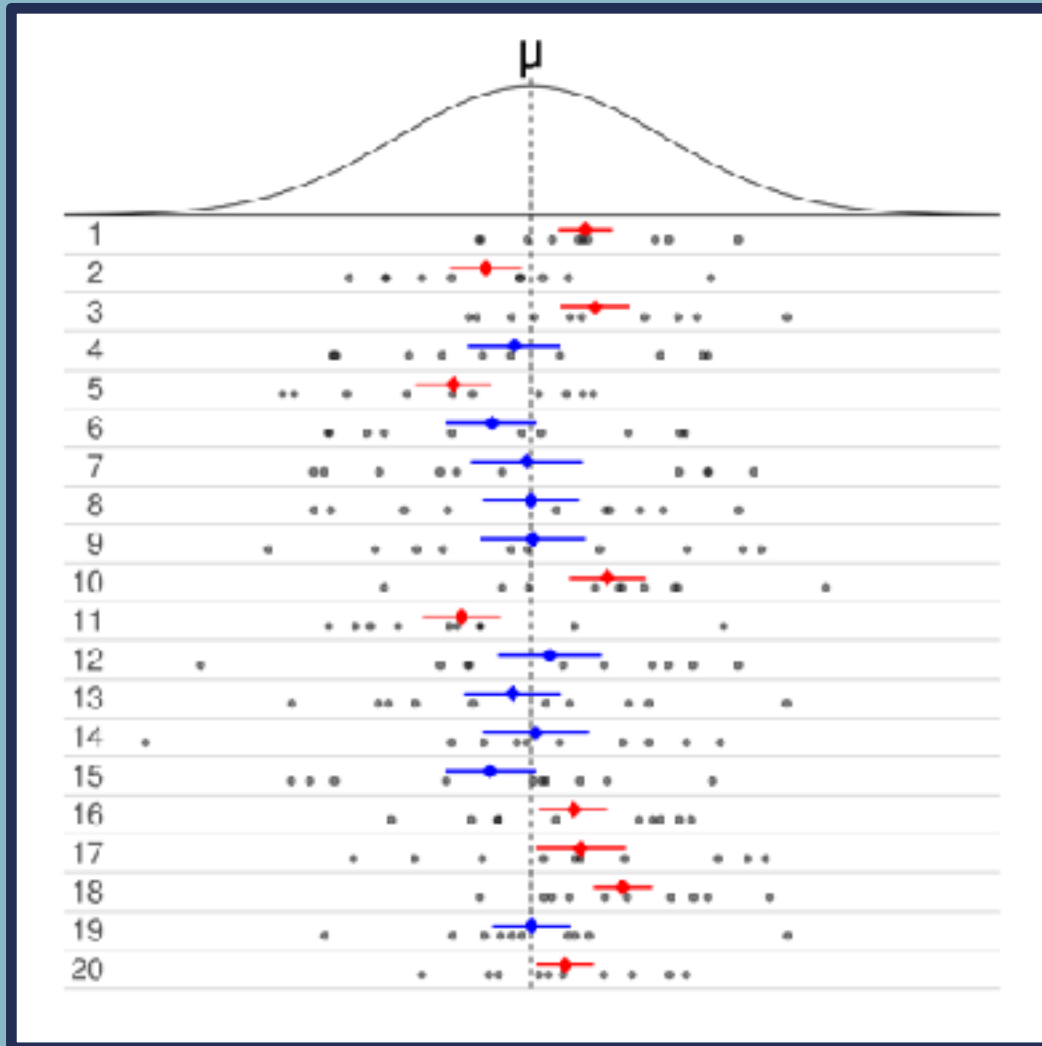
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- $U(\theta) = \nabla \mathcal{L}(\theta) = \nabla p(D | \theta) = \left(\frac{\delta \mathcal{L}(\theta)}{\delta \theta_1}, \dots, \frac{\delta \mathcal{L}(\theta)}{\delta \theta_K} \right)$ for k parameters in vector θ

In Practice



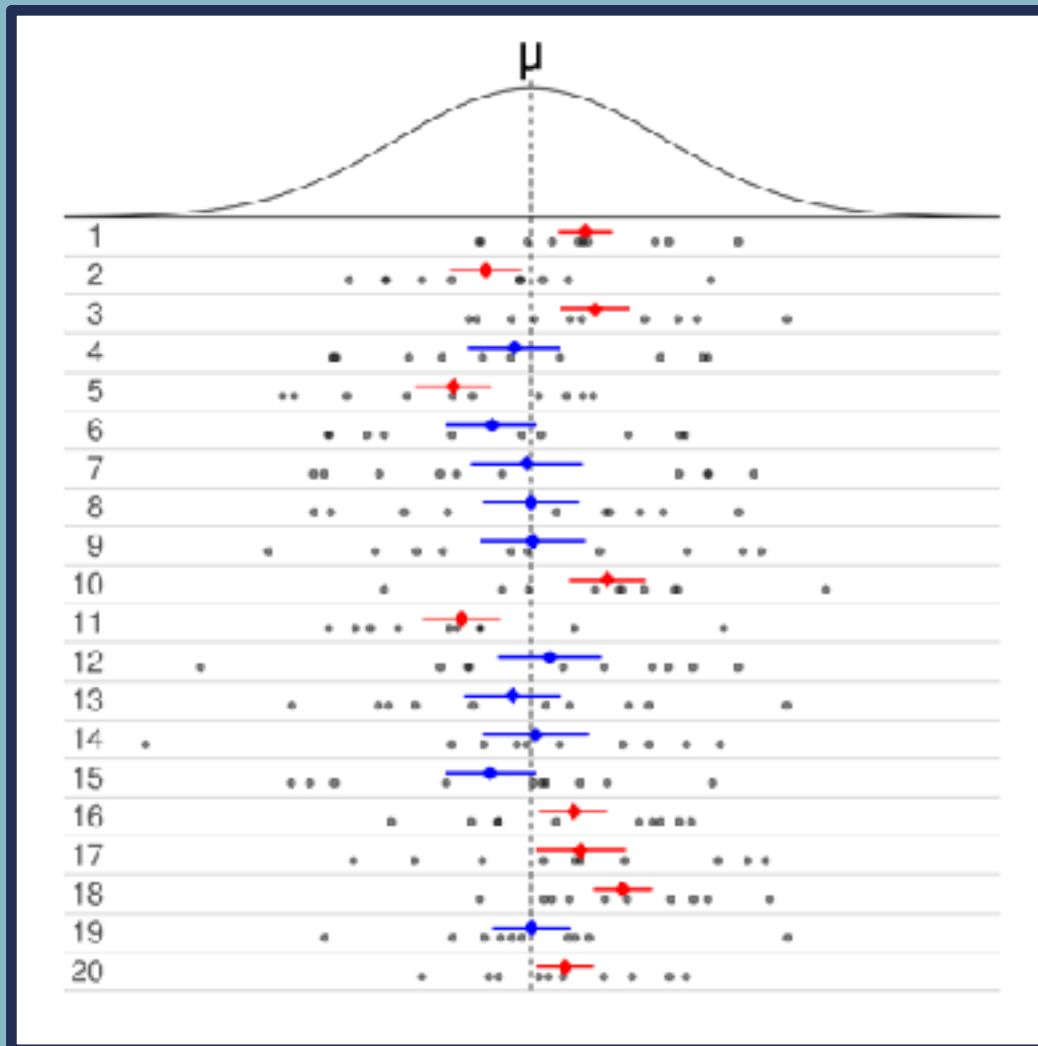
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- Use (stochastic) gradient descent to find the minimum if you have analytical gradients, or algorithms that numerically approximate the gradient otherwise

Confidence Intervals



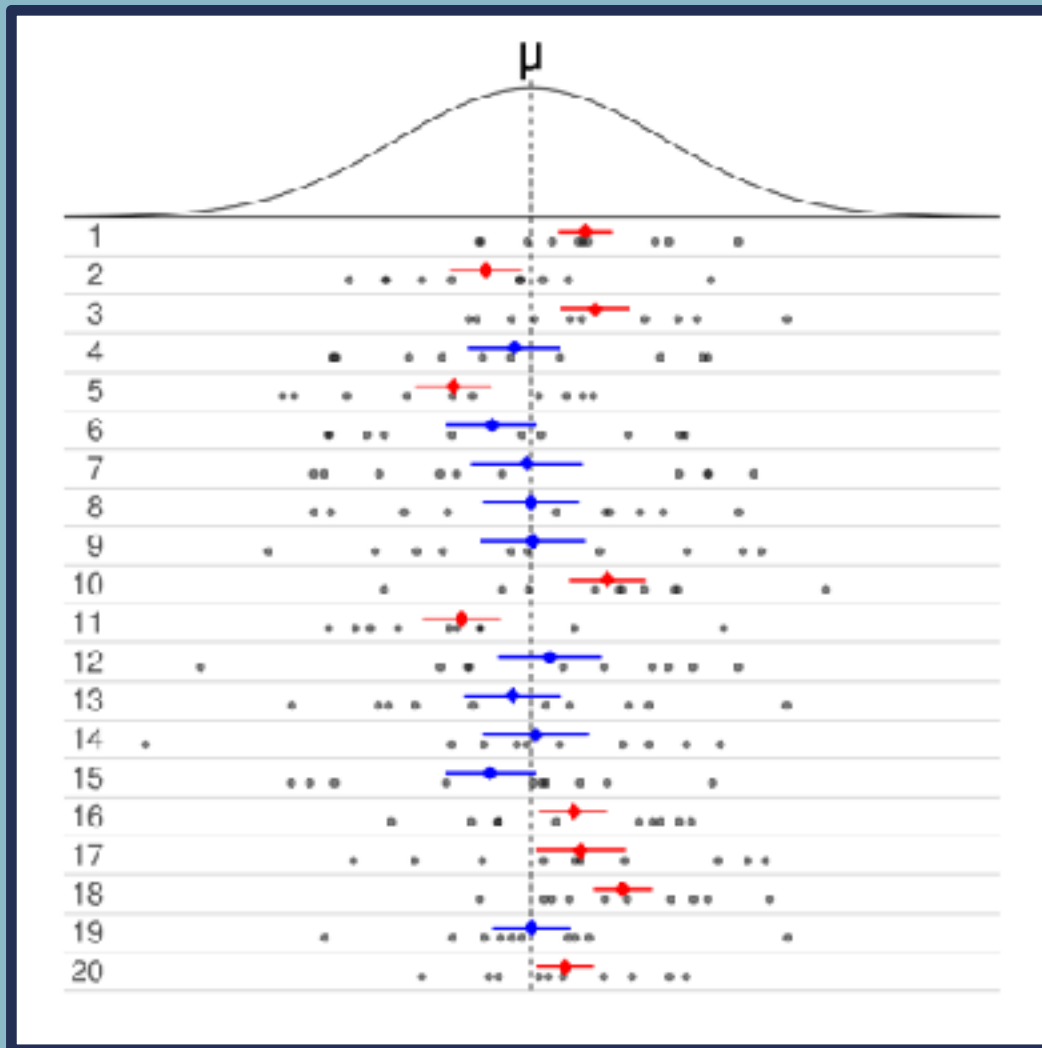
Confidence Intervals

- Range of values used to estimate an unknown statistical parameters



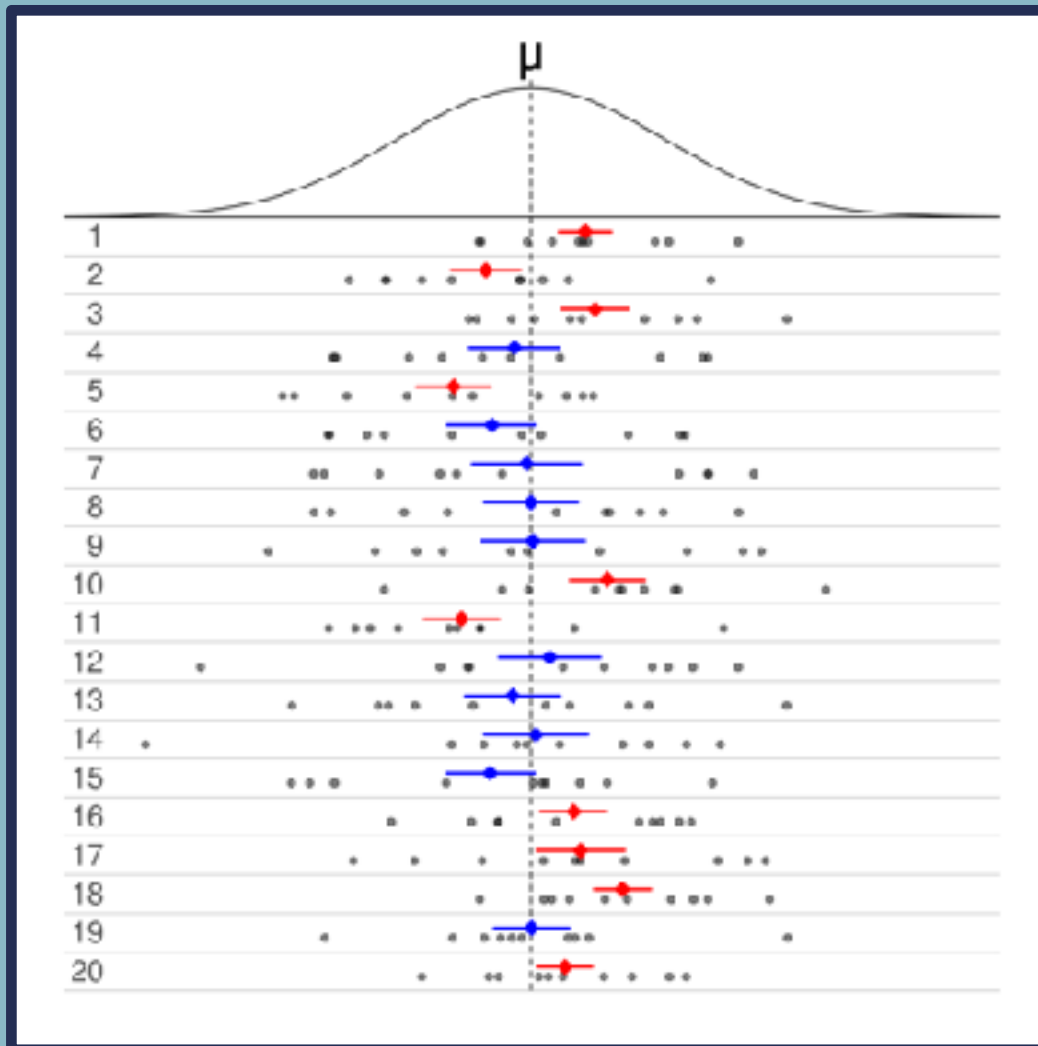
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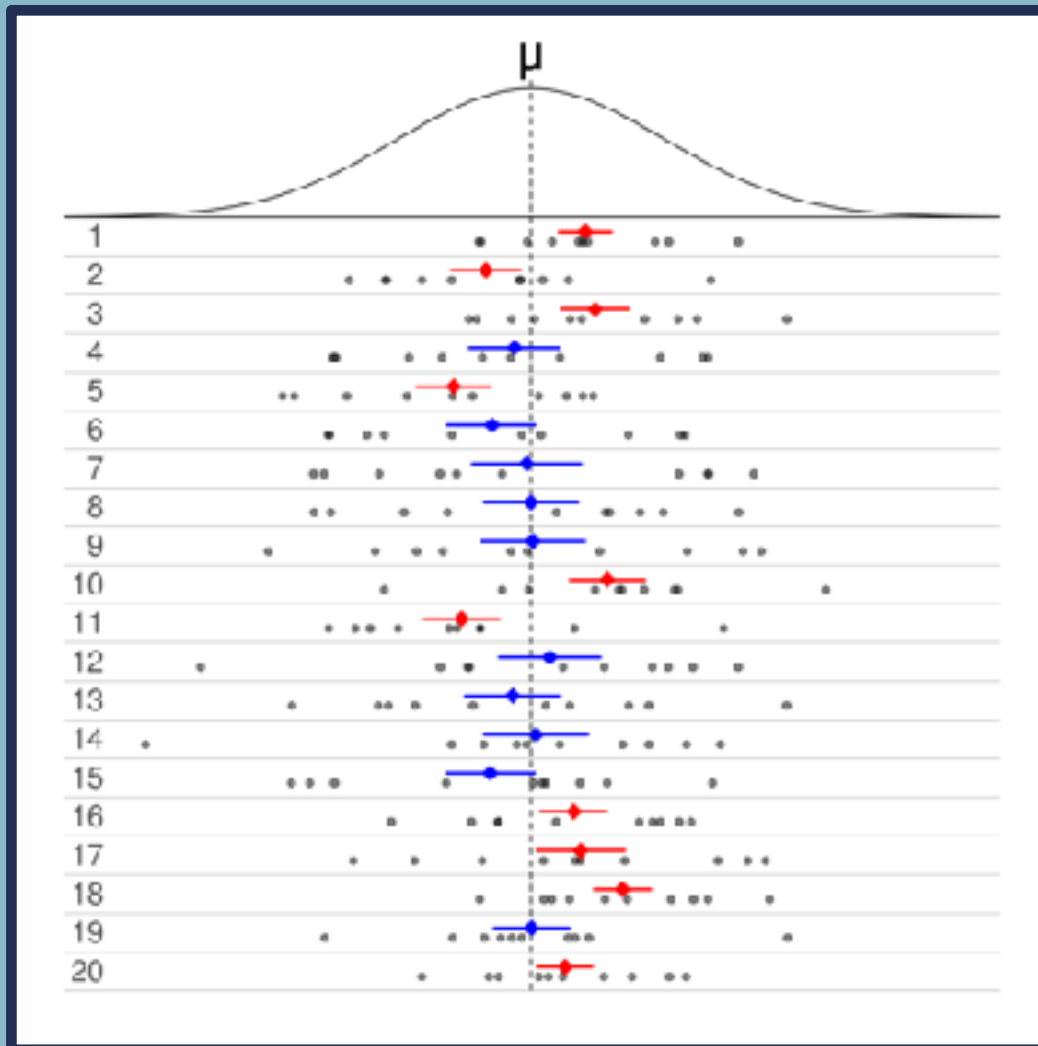


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- Specify confidence level (usually $68\% = 1\sigma$, $95\% = 2\sigma$)
- Does not imply a 95% probability that the true parameter lies within a particular calculated interval!
- If the same sampling procedure were repeated 100 times from the same population, approximately 95 of the resulting intervals would be expected to contain the true population mean